MONOTONE NONPLANAR SOLUTIONS OF THE FRACTIONAL ALLEN–CAHN EQUATION IN $\mathbb{R}^9$

ABSTRACT. For each $0 < s < 1/2$ we construct a bounded entire solution of $(-\Delta)^s U = U - U^3$ in $\mathbb{R}^9$ that is strictly increasing in its ninth variable and depends on more than one linear coordinate. The geometric input is the strict fractional mean curvature inequality for an explicit cap inside a chamber of the eight-dimensional Simons cone. Transport of the diffuse saddle to dilates of this cap gives ordered barriers and a reflection-symmetric minimizer in dimension eight. An infinite-cylinder comparison and a centered slope selection then produce the nine-dimensional entire minimizer.

Draft status. This candidate proof was obtained by our team on November 05, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction

Consider the cubic fractional Allen–Cahn equation

$$(-\Delta)^s U = U - U^3 \qquad \text{in } \mathbb{R}^n, \qquad 0 < s < 1, \tag{1.1}$$

where the fractional Laplacian has Fourier symbol $|\xi|^{2s}$. The fractional De Giorgi question asks when a bounded solution with a strictly positive derivative in one direction must be a planar layer. Here a solution is planar if $U(X) = p(e \cdot X)$ for some unit vector e and some one-variable function p.

Theorem 1. *For every $0 < s < 1/2$, equation (1.1) in $\mathbb{R}^9$ has a smooth solution $U : \mathbb{R}^9 \to (-1, 1)$ with $\partial_{X_9} U > 0$ everywhere. It is an entire minimizer of the fractional Allen–Cahn energy and is nonplanar.*

An entire minimizer has finite relative energy (2.1) on each bounded Lipschitz open set and minimizes it among competitors that agree with the solution almost everywhere outside that set.

For the classical Allen–Cahn equation, del Pino, Kowalczyk, and Wei constructed a nonplanar monotone entire solution in dimension nine by placing a transition layer near a dilation of the nonaffine minimal graph of Bombieri, De Giorgi, and Giusti over $\mathbb{R}^8$ [1, 8]. Jerison and Monneau gave a variational route based on a nonconstant entire minimizer in one lower dimension that is invariant under reflection of each coordinate [14, Theorem 1.7]. Liu, Wang, and Wei constructed such minimizers for the classical equation and obtained another proof of the dimension-nine counterexample [15, Corollary 3 in the revised author manuscript]. The construction here takes the variational route.

The rescaled fractional Allen–Cahn energies Γ-converge to a multiple of the nonlocal perimeter of order $2s$ for $0 < s < 1/2$, and to a multiple of the classical perimeter for $1/2 \le s < 1$, with the appropriate scaling in each case [18, Theorem 1.4]. Below $s = 1/2$, this links the diffuse transition to the nonlocal interface theory of Caffarelli, Roquejoffre, and Savin [5].

Sire and Valdinoci proved that bounded monotone solutions in dimension two are one-dimensional for every $0 < s < 1$ [21, Theorem 3]. In dimension three, Dipierro, Farina, and Valdinoci treated $0 < s < 1/2$ [10, Theorem 1.1], and Cabré and Cinti treated $1/2 \le s < 1$ [2, Theorem 1.5]. Cabré, Cinti, and Serra proved sharp energy and BV estimates for stable solutions in the nonlocal perimeter regime and showed that subsequential blow-downs give cones weakly stable for the nonlocal perimeter. They obtained one-dimensional symmetry for monotone solutions in dimension four for s sufficiently close to $1/2$ from below [4, Theorems 2.1, 2.6, and 2.10 and Corollary 1.11]. Figalli and Serra proved the dimension-four result at $s = 1/2$ [13, Corollary 1.6].

For s sufficiently close to $1/2$ from below, Dipierro, Serra, and Valdinoci proved one-dimensional symmetry for entire minimizers through dimension seven and for strictly monotone solutions through dimension eight with pointwise limits ± 1 at the two ends of the monotone direction [11]. Savin

proved the corresponding statements for $1/2 \le s < 1$ [19, 20, Theorems 1.2–1.3]. On the construction side, Chan, Liu, and Wei proved the existence of unstable catenoidal solutions in $\mathbb{R}^3$ for $1/2 < s < 1$. Their preprint also announces a construction with Dávila and del Pino of monotone nonplanar solutions in dimensions $n \ge 9$ in that same exponent range [6, Theorems 1.3–1.4 and the discussion that follows].

The GitHub collection [27] lists five rigidity drafts as candidates under verification. Two address dimension four for $0 < s < 1/2$ [22, Theorem 1.2] and $1/2 < s < 1$ [23, Theorem 1.1]. Two address dimension eight for s near 1 [25, Theorem 1.1] and for s near $1/2$ from below [24, Theorem 1.1], without prescribing the limits in the monotone direction. The fifth gives a conditional argument through dimension eight at a fixed $s \in (1/2, 1)$. It assumes a sharp BV growth bound, uniform in the center of the ball, for every bounded stable classical solution in each dimension up to seven at that exponent [26, Theorem 1.1].

The geometric input for Theorem 1 comes from the draft on entire nonlocal minimal graphs [29, Lemma 5.4 and Theorem 4.9]. We use the geometry of an explicit cap inside one chamber of the eight-dimensional Simons cone and the strictly negative fractional mean curvature of its boundary in the exterior-minus-interior convention. The curvature bound quantifies the negative margin as the cap approaches the cone at infinity. We transfer this sign to a diffuse transition of fixed thickness by transporting the exact fractional saddle from the cone to dilates of the cap. The nonlocal operator sees the change of profile at every distance. Since the curvature decays along the cap, the residual estimate must control the transport error on the same spatial scale. The use of ordered diffuse barriers is related to the p-Laplace construction in [28, Theorem 6.7].

Section 2 establishes the variational comparison and compactness tools and shows that the increasing one-dimensional layer q, extended to higher dimensions, is an entire minimizer. Section 3 states the cap estimates in the coordinates used for transport, and Section 4 proves the saddle bounds needed to control its variation along the cone. In Section 5 we preserve signed normal distance and use arclength in the plane of the two block radii. The leading change of the kernel in the angular directions integrates to zero against the profile frozen in those directions. The resulting expansion has the cap curvature as its leading term, with a smaller error uniformly on the exterior portion of each fixed-width band about the dilated cap. Section 6 joins the exterior profile to a normal profile inside a ball whose radius is proportional to the dilation scale. It also corrects the residual where the profile is close to the wells. Sliding the resulting barriers against ball minimizers and sending the ball radius to infinity gives the nonconstant, coordinatewise reflection-symmetric entire minimizer v of Proposition 7.

Section 7 then works on cylinders with bounded horizontal base and interpolates between the lateral data $q(z)$ and $v(X)$, where X is horizontal and z is vertical. The finite interaction across horizontal cuts gives control of the

vertical end profiles and yields uniqueness among the nondecreasing cylinder minimizers. After vertical translation to fix the central value $c = v(0)$, the derivative at the center varies continuously. It equals $q'(q^{-1}(c))$ for the planar datum and tends to zero as the lateral datum approaches $v(X)$. In Theorem 8, we select an intermediate derivative before sending the base radius to infinity. The horizontal reflections and strict vertical monotonicity force any planar limit to be a vertical translate of q, whose derivative at value c is $q'(q^{-1}(c))$. The selected derivative excludes this possibility.

2. Energy and comparison

Throughout the paper $0 < s < 1/2$, $\sigma = 2s$, and $L = (-\Delta)^s$ has Fourier symbol $|\xi|^{2s}$. In dimension n its kernel is $K_n(h) = c_{n,s}|h|^{-n-\sigma}$. Set

$$f(a) = a - a^3, \qquad W(a) = \frac{(1-a^2)^2}{4}, \qquad Q_A = (\mathbb{R}^n)^2 \setminus (A^c \times A^c).$$

For a bounded Lipschitz open set $A \subset \mathbb{R}^n$, define

$$\mathcal{E}_n(u; A) = \frac{1}{4} \iint_{Q_A} (u(P) - u(Q))^2 K_n(P - Q)\, dP\, dQ + \int_A W(u). \tag{2.1}$$

The pair domain consists of $A \times A$ and both orders of the cross pairs in $A \times A^c$. Exterior data are prescribed almost everywhere on A^c. An entire minimizer has finite relative energy on every bounded Lipschitz open set A and minimizes it among measurable competitors with the same exterior values.

Lemma 2. *If A is a bounded Lipschitz open set, then $\int_A \int_{A^c} K_n(P - Q)\, dQ\, dP < \infty$. For exterior data in $[-1, 1]$, clipping to this interval fixes the data and does not increase the energy. There is a minimizer on A, and every minimizer takes values in $[-1, 1]$ almost everywhere. In A it has a smooth representative that solves $Lu = f(u)$ pointwise. More generally, bounded distributional solutions of this equation have uniform interior estimates of every order on compact subsets. For fixed dimension, exponent, derivative order and distance to the boundary, the estimates are uniform under a common global bound on the solutions. For any bounded measurable function locally Lipschitz in A, the absolutely convergent kernel integral represents its distributional fractional Laplacian in A.*

Minimality on A restricts to every bounded Lipschitz open subset of A. It also passes to almost-everywhere limits on $\mathbb{R}^n$ of $[-1, 1]$-valued minimizers on A, including sequences with varying exterior data.

Proof. Cross interaction and clipping. If $P \in A$ and $P + h \notin A$, the segment from P to $P + h$ meets ∂A, so $\operatorname{dist}(P, \partial A) \le |h|$. For $0 < r < 1$, the r-neighborhood of ∂A has volume at most $C_A r$. To see this, cover the compact boundary by finitely many shrunken coordinate cylinders whose closures lie in Lipschitz graph charts. For r smaller than the minimum chart margin, a nearest boundary point to any point at distance less than r

lies in a shrunken cylinder, and the nearby point is still in the original chart. If the chart is $x_n = \psi(x')$, the vertical distance of the nearby point from the graph at its own horizontal coordinate is at most $(1 + \operatorname{Lip}\psi)r$. This bounds the volume by the sum of the band volumes over the bounded chart bases. For the remaining $r < 1$, enlarge C_A using the finite volume of the unit neighborhood of ∂A. Using the bound $|A|$ for displacements of length at least one, we obtain

$$\int_A \int_{A^c} K_n(P-Q)\,dQ\,dP = \int_{\mathbb{R}^n} K_n(h)|\{P \in A : P + h \in A^c\}|\,dh$$
$$\leq C_A \int_0^1 r^{-\sigma}\,dr + C|A| \int_1^\infty r^{-1-\sigma}\,dr < \infty.$$

With exterior data in $[-1,1]$, the competitor equal to zero in A has finite energy by this estimate. Projection onto $[-1,1]$ fixes the datum and decreases the kinetic integrand by the 1-Lipschitz property of the projection. On the set where $|u| > 1$ it strictly decreases $W(u)$ to zero. Consequently the infimum can be computed in $[-1,1]$, and any minimizer already lies in that interval almost everywhere.

Existence and closure. We first give the limit comparison used in both claims. Suppose $u_j : \mathbb{R}^n \to [-1,1]$ converge almost everywhere to u, and each u_j minimizes on A up to an energy error $\varepsilon_j \to 0$ among competitors with its exterior datum. For any finite-energy competitor b for u on A, we may clip b to $[-1,1]$ by the preceding argument. Define $b_j = b$ on A and $b_j = u_j$ on A^c. The potential and the $A \times A$ term of b_j are those of b. Its cross term converges to that of b by dominated convergence, using the bound $4K_n(P-Q)$ on the cross integrand and the finite cross interaction. Fatou's lemma and the comparison with b_j give

$$\mathcal{E}_n(u;A) \leq \liminf_{j\to\infty} \mathcal{E}_n(u_j;A) \leq \lim_{j\to\infty} \big(\mathcal{E}_n(b_j;A) + \varepsilon_j\big) = \mathcal{E}_n(b;A).$$

A finite-energy b exists by the constant interior construction, so this proves minimality and finiteness for the limit.

For existence, take a minimizing sequence with the fixed exterior datum and with values in $[-1,1]$. Its $L^2(A)$ norms are bounded, and the $A \times A$ energy bounds their Gagliardo seminorms in $W^{s,2}(A)$. Theorem 5.4 of [9] gives the extension property for a Lipschitz domain, and their Theorem 7.1 with both Lebesgue exponents equal to two gives convergence of a subsequence in $L^2(A)$. Extract further for almost-everywhere convergence on A. Extend the limit by the fixed datum and apply the limit comparison. Taking $\varepsilon_j = 0$ in that comparison also proves closure for minimizers with varying data.

For restriction, let $A' \subset A$ be bounded and Lipschitz. A finite-energy competitor on A' agrees with u on $(A')^c$, so its pair integrands on $Q_A \setminus Q_{A'}$ and its potential on $A \setminus A'$ agree with those of u. These unchanged contributions are finite since u has finite energy on A. Cancelling them in the comparison on A gives the comparison on A'.

First variation. For $\varphi \in C_c^\infty(A)$, the test increment has finite squared integral against the kernel on Q_A. Indeed at least one endpoint lies in A, and the squared increment is bounded by $C\min\{|P-Q|^2, 1\}$. Integration in the displacement followed by integration over A gives finiteness. Cauchy–Schwarz gives integrability of the mixed term with the minimizer u. Since minimization holds against unrestricted competitors, we can differentiate the energy of $u+\varepsilon\varphi$ at zero. For the potential, when $|\varepsilon| \le 1$ the arguments of W stay in a fixed compact interval. The mean value theorem then bounds the potential difference quotients by $C|\varphi|$, so dominated convergence applies. Expanding the kinetic square and using $W' = -f$ gives

$$\frac{1}{2}\iint_{\mathbb{R}^n\times\mathbb{R}^n} (u(P)-u(Q))(\varphi(P)-\varphi(Q))K_n(P-Q)\,dP\,dQ = \int_A f(u)\varphi. \tag{2.2}$$

Both variables may range over $\mathbb{R}^n$, since the test difference vanishes when both lie in A^c. Truncate the integral to $|P-Q| > \delta$ and interchange P, Q in the term containing $-\varphi(Q)$. By symmetry the truncated left-hand side is $\int_{\mathbb{R}^n} uL_\delta\varphi$, where

$$L_\delta\varphi(P) = \int_{|P-Q|>\delta} \big(\varphi(P)-\varphi(Q)\big)K_n(P-Q)\,dQ.$$

The truncated pair integral tends to the full one by absolute integrability. Also $L_\delta\varphi \to L\varphi$ in L^1: at short distances use $\|\varphi(\cdot+h)-\varphi\|_{L^1} \le |h|\|\nabla\varphi\|_{L^1}$ and $\sigma < 1$, and at large distances use the bound $2\|\varphi\|_{L^1}$. Thus (2.2) gives $Lu = f(u)$ in distributions.

Interior regularity. The bounded-source estimate and the interior Schauder gain of order σ are given by [17, Theorem 1.1(a) and Corollary 3.5]. By evenness, the generator $-L$ has the second-increment form of that reference. Its spectral measure μ is a positive constant multiple of surface measure on S^{n-1}, as is seen by writing the kernel in polar coordinates. The mass is finite. For every unit ξ, the directional integral $\int_{S^{n-1}} |\xi\cdot\theta|^\sigma\,d\mu(\theta)$ is the same strictly positive number by rotation invariance. This gives the required uniform ellipticity, and the constant density is Hölder for every exponent in $(0,1)$. The source for the generator $-L$ is $-f(u)$. These estimates apply on interior balls by translation and rescaling. The weak formulation of [17, Definition 2.2] is the distributional one above for bounded functions, which satisfy the growth condition in that definition.

Here are the details of the iteration. Choose $0 < b < \sigma$ so that jb and $jb+\sigma$ are nonintegers for all integers $j \ge 1$. This excludes only the countable set of values k/j and $(k-\sigma)/j$, with $k \in \mathbb{Z}$ and $j \ge 1$. The bounded-source estimate gives C^σ, and hence C^b, on a smaller ball. Suppose that u is bounded in C^{jb} on an interior ball. If $jb < 1$, the cubic gives $f(u)$ bounded in C^{jb} there. The Schauder gain gives $C^{jb+\sigma}$ on a smaller ball, from which we retain $C^{(j+1)b}$.

If $jb = k + a$ with $k \ge 1$ and $0 < a < 1$, choose a smooth cutoff χ supported in the ball of the $C^{k,a}$ estimate and equal to one on a smaller ball.

On a ball compactly contained in $\{\chi = 1\}$, the cutoff equation is

$$L(\chi u)(P) = f(u(P)) + \int_{\mathbb{R}^n} (1 - \chi(Q))u(Q)K_n(P - Q)\, dQ.$$

Indeed, subtract $L((1-\chi)u)$ from Lu in distributions. For a test supported in the smaller ball, the test function itself vanishes on the support of $(1-\chi)u$. On that support its fractional Laplacian is the negative integral of the test values against the kernel. Separation of the supports and the integrable tail justify interchanging the integrals, giving the displayed positive tail term. The same separation makes the kernel derivatives in P integrable, with bounds controlled by the separation and $\|u\|_\infty$, so the term is smooth. Let β be a multi-index of order k. For a compactly supported smooth test φ, differentiating its kernel increments gives $D^\beta L\varphi = LD^\beta\varphi$. This is justified by Lipschitz bounds for each differentiated increment at short distances and boundedness in the tail. Put $w = \chi u$. Since w and $D^\beta w$ have compact support, integration by parts against $L\varphi$ is valid after multiplying it by a cutoff equal to one near their support. Thus

$$\langle LD^\beta w, \varphi\rangle = (-1)^{|\beta|}\langle w, D^\beta L\varphi\rangle = \langle D^\beta Lw, \varphi\rangle.$$

The function $D^\beta(\chi u)$ is bounded globally by the product rule. Each product uses a derivative of u controlled on the ball containing the support and a derivative of the cutoff. Its source is $D^\beta f(u)$ plus a derivative of the smooth integral. The product rule bounds $D^\beta f(u)$ in C^a, since every factor involves derivatives of u of order at most k. Applying the Schauder gain to each $D^\beta(\chi u)$ gives $C^{k+a+\sigma}$ for u on a smaller ball. Retaining order $(j+1)b$ in either case allows the iteration to continue. For any prescribed order, choose the number of steps in advance. Use nested concentric balls with fixed positive gaps between their radii, from the initial interior radius to half that radius, reserving intermediate radii for the cutoffs. Cutoffs obtained by rescaling a fixed smooth function have derivatives bounded in terms of these gaps and the derivative orders. This yields uniform bounds on the smaller balls, hence on compact interior sets, with the stated dependence on the distance to the boundary.

Finally, for any bounded measurable u locally Lipschitz in A, the local Lipschitz bound and $\sigma < 1$ imply absolute convergence of the pointwise integral for Lu in the interior. The contribution from $|P - Q| < \delta$ is $O(\delta^{1-\sigma})$ locally uniformly. For fixed δ, the truncated integral is continuous there because u is continuous there and convolution of bounded u with the truncated L^1 kernel is continuous by continuity of translations in L^1. On interior tests, the symmetry identity $\int (L_\delta u)\varphi = \int uL_\delta\varphi$ passes to the limit by the local uniform convergence on the left and the L^1 estimate on the right. Thus the pointwise integral represents the distributional operator. For the smooth representative of a solution, the two sides of the equation are continuous, so the equation holds at every interior point. □

Lemma 3. *If $w : \mathbb{R}^n \to \mathbb{R}$ is bounded and nonnegative almost everywhere, is Lipschitz near P, and satisfies $w(P) = Lw(P) = 0$, then $w = 0$ almost everywhere.*

Let a, b be minimizers on the same bounded Lipschitz open set with exterior values in $[-1, 1]$. If their exterior data agree, then a and b are comparable almost everywhere. If the exterior data satisfy $a \le b$ almost everywhere and $a < b$ on a set of positive measure, then $a < b$ at every interior point for the smooth representatives. An entire minimizer with values in $[-1, 1]$ is the unique minimizer, up to almost-everywhere equality, on any ball with its own exterior datum.

Proof. At the stated zero of w, local Lipschitz continuity and $\sigma < 1$ give absolute convergence near P, while boundedness of w gives convergence against the integrable kernel tail. Thus

$$0 = Lw(P) = -\int_{\mathbb{R}^n} w(Q)K_n(P - Q)\,dQ.$$

Since $w \ge 0$ and the kernel is positive, $w = 0$ almost everywhere.

For each of a, b, the competitor equal to zero on A and to the prescribed values outside has finite energy by Lemma 2. Thus both minimizing energies are finite. Use the smooth representatives in A. They lie in $[-1, 1]$ there by continuity and the almost-everywhere bound. Choose exterior representatives in the same interval by modifying them on null sets. This leaves the energies and all pointwise operator integrals inside A unchanged.

Taking the pointwise minimum and maximum preserves the potential sum. At pairs with the same order the kinetic sum is unchanged as well. If $a(P) > b(P)$ and $a(Q) < b(Q)$, the decrease in the sum of the two squared increments is

$$\begin{aligned} &(a(P) - a(Q))^2 + (b(P) - b(Q))^2 \\ &\quad - (b(P) - a(Q))^2 - (a(P) - b(Q))^2 \\ &= 2(a(P) - b(P))(b(Q) - a(Q)) > 0. \end{aligned} \tag{2.3}$$

The other opposite order follows by interchanging P and Q. When $a \le b$ on the exterior, the minimum has the datum of a and the maximum has the datum of b. The pairwise inequality also shows they have finite energy. If the exterior data agree, positive-measure sets with both strict orders would lie in A, and their product would give a strict energy decrease. If the exterior data are strictly ordered on a positive-measure set, a reversed interior order set of positive measure would give the same strict decrease when paired with that exterior set. Both contradict the sum of the two minimizing inequalities.

For strictness in the interior, continuity upgrades the almost-everywhere order to $a \le b$ at every interior point. Let $w = b - a$. At any interior point P where the smooth representatives touch, $w(P) = 0$ and $Lw(P) = f(b(P)) - f(a(P)) = 0$. The contact assertion then gives $w = 0$ almost everywhere, contrary to the strict exterior order.

Now let v be an entire minimizer and w a minimizer on B_R with exterior datum v. Entire minimality gives the same minimum energy for v and w on B_R. The pair domain $Q_{B_{R+1}} \setminus Q_{B_R}$ contains only pairs with both points outside B_R, so the kinetic integrals there agree for v and w. The potential integrals on $B_{R+1} \setminus B_R$ also agree. These added integrals are finite because v has finite energy on B_{R+1}. Adding them to the equal energies on B_R gives equal energies on B_{R+1}, with the same exterior datum there. Thus v, w are both minimizers on B_{R+1} and are comparable by the equal-datum case. By continuity they agree at every point of $B_{R+1} \setminus \overline{B}_R$, where they agree almost everywhere by the original exterior condition. Apply the contact argument at such a point to the nonnegative difference, in the order given by comparability. This proves uniqueness. □

Kernel marginals and the planar minimizer. For a coordinate split $h = (\xi, \zeta) \in \mathbb{R}^k \times \mathbb{R}^{n-k}$, $1 \le k < n$, the normalized kernel satisfies

$$\int_{\mathbb{R}^{n-k}} K_n(\xi, \zeta)\, d\zeta = K_k(\xi) \qquad (\xi \neq 0). \tag{2.4}$$

In fact, scaling ζ by $|\xi|$ gives a positive constant times $|\xi|^{-k-\sigma}$. To identify that constant, evaluate the Fourier symbol at the unit vector in the first retained coordinate. In dimension j the normalization reads

$$\int_{\mathbb{R}^j} (1 - \cos h_1) K_j(h)\, dh = 1.$$

The cosine appears by evenness of the kernel, and the integral converges by the quadratic increment at zero and the integrable tail. Tonelli's theorem applied to the same identity in dimension n fixes the constant in the marginal to be $c_{k,s}$. In particular, L applied to a bounded smooth function of one coordinate reduces to the one-dimensional L, by (2.4) and absolute convergence for $\sigma < 1$.

By [3, Theorem 2.4] there is a one-dimensional layer solving $Lq = f(q)$ with $q' > 0$ and $q(\pm\infty) = \pm 1$, unique up to translation in that class. Indeed $W > W(-1) = W(1) = 0$ on $(-1, 1)$, $f(\pm 1) = 0$, and $f'(\pm 1) = -2$. A smooth cutoff of f equal to the cubic near $[-1, 1]$ gives the required global Hölder bound for f' with any exponent in $(1 - \sigma, 1)$, and the cutoff leaves the layer equation unchanged. The end limits and strict monotonicity give a unique zero. Translate the profile so that $q(0) = 0$.

The interior estimates on translated unit intervals give uniform bounds on the derivatives of q. For $n \ge 2$, set $U(P) = q(P_n)$. The kernel marginal gives $LU = f(U)$, and U inherits those derivative bounds. In particular, $U \in C^{1+\sigma+\gamma}(\mathbb{R}^n)$ for any $0 < \gamma < 1 - \sigma$, since the bounded first derivatives are globally Lipschitz. Also $\partial_{P_n} U = q'(P_n) > 0$ and $U(P', P_n) \to \pm 1$ as $P_n \to \pm\infty$ for every P'.

These are the hypotheses of [7, Theorem 3] for the kernel $K = K_n$ and potential W. The kernel is even and satisfies their two-sided bounds of order $n + \sigma$, and W is smooth, nonnegative, and satisfies $W(\pm 1) = W'(\pm 1) = 0$.

Their energy uses $1/4$ of the ordered pair integral on Q_A, as in (2.1). Their operator L_K equals $-L$ for this kernel, so their equation $L_K U = W'(U)$ is exactly $LU = f(U)$. Their conclusion gives minimality on every ball. For a bounded Lipschitz open set A, take a ball containing A and use the restriction part of Lemma 2.

3. The cap and its orientation

For $P = (x, y) \in \mathbb{R}^4 \times \mathbb{R}^4$, write $a = |x|$, $b = |y|$, and set

$$D = \{b > a\}, \qquad \Sigma = \partial D, \qquad \rho = \frac{a+b}{\sqrt{2}}, \qquad t = \frac{b-a}{\sqrt{2}}.$$

By continuity, ∂D lies in $\{b = a\}$. Conversely, at any point with $b = a > 0$, slightly enlarging either block gives points on the two sides. At the origin use small vectors in the two factor spaces separately. Thus $\Sigma = \{b = a\}$. The region of possible (ρ, t) is $\rho \geq |t|$, and $|P|^2 = \rho^2 + t^2$, so $|P|/\sqrt{2} \leq \rho \leq |P|$. The map $P \mapsto (a, b)$ is 1-Lipschitz by the reverse triangle inequality in each block. Its two unit linear projections ρ and t are therefore 1-Lipschitz as well. For any fixed block radii $c, e \geq 0$, the reverse triangle inequality bounds the squared distance from P to a point with those radii below by $(a-c)^2+(b-e)^2$. Equality is attained by matching the direction of each nonzero block of P, with arbitrary direction for a zero block. On Σ the two radii coincide, so we minimize over their common value $u \geq 0$. The resulting squared distance at fixed u is

$$(a-u)^2 + (b-u)^2 = 2\left(u - \frac{a+b}{2}\right)^2 + \frac{(b-a)^2}{2}.$$

Thus t is the signed Euclidean distance to Σ, positive on D.

For a set G, we use the exterior-minus-interior convention

$$H_\sigma[G](p) = \lim_{\varepsilon \downarrow 0} \int_{|Q-p|>\varepsilon} \frac{\mathbf{1}_{G^c}(Q) - \mathbf{1}_G(Q)}{|p-Q|^{8+\sigma}}\, dQ$$

when the limit exists. At a point $p = (x_0, y_0) \in \Sigma \setminus \{0\}$, choose an orthogonal map R of $\mathbb{R}^4$ with $Rx_0 = y_0$. The transformation $(x, y) \mapsto (R^{-1}y, Rx)$ is an isometry that fixes p and interchanges the two open chambers. Under this change of variables the kernel and every centered truncation are unchanged, while $\mathbf{1}_{D^c} - \mathbf{1}_D$ changes sign away from Σ. The cone has measure zero by Fubini, since for each fixed x the section in the y-block is the sphere of radius $|x|$. For every $\varepsilon > 0$ the truncated integrand is absolutely integrable, with integral in absolute value bounded by $|S^7| \int_\varepsilon^\infty r^{-1-\sigma}\, dr$. The change of variables therefore gives zero at every truncation, and $H_\sigma[D](p) = 0$.

The cap is

$$h(r) = (1 + 2r^2/5)^{-3/8}, \qquad E = \{|y|^2 - |x|^2 > h(2|x||y|)\}. \tag{3.1}$$

By the geometry lemma in [29, Lemma 5.4], its smooth boundary is the graph $b = h_*(a)$. Here $h_*(0) = 1$, $h_*(a) > a$, and $0 < h_*'(a) < 1$ for $a > 0$. The function h_* is smooth and even at zero. At $a = 0$ the x-block

angle collapses, and the Cartesian boundary is smooth there. Indeed, the defining function in (3.1) depends smoothly on a^2, b^2 and has y-gradient $2y$ at $x = 0,\ |y| = 1$.

Along the boundary, $\rho = (a + h_*(a))/\sqrt{2}$ increases from $\rho_{\mathrm{ax}} = 1/\sqrt{2}$ to infinity. For $a > 0$ its derivative is $(1 + h_*'(a))/\sqrt{2} > 0$, so the inverse is smooth by the inverse function theorem. Thus the boundary is $t = \eta(\rho)$ in these coordinates, with $\eta(\rho_{\mathrm{ax}}) = \rho_{\mathrm{ax}}$ and $\eta(\rho) > 0$. Since $t = (h_*(a) - a)/\sqrt{2}$ on the boundary,

$$\eta'(\rho) = \frac{h_*'(a) - 1}{h_*'(a) + 1} < 0 \qquad (\rho > \rho_{\mathrm{ax}}).$$

At fixed ρ, increasing t increases b and decreases a, so $b - h_*(a)$ increases strictly. For each fixed a, the function $b \mapsto b^2 - a^2 - h(2ab)$ is strictly increasing for $b \geq 0$, because h is nonincreasing. It vanishes at the boundary radius $h_*(a)$ by continuity. Thus E is the side $b > h_*(a)$, or equivalently $t > \eta(\rho)$ in the admissible region with $\rho > \rho_{\mathrm{ax}}$. For $\rho \leq \rho_{\mathrm{ax}}$, $b \leq \sqrt{2}\rho \leq 1 \leq h_*(a)$, so there are no points of E there. In the plane obtained by fixing the two block directions, the inward unit normal to the graph at $\rho > \rho_{\mathrm{ax}}$ is

$$\nu = \frac{(-\eta'(\rho), 1)}{\sqrt{1 + \eta'(\rho)^2}}.$$

This is the direction of increase of the defining function $t - \eta(\rho)$. Invariance under block rotations makes the normal orthogonal to the angular directions. On $\overline{E} \cap \overline{B}_R$ for $R > 0$, the defining inequality gives $2\rho t \geq h(\rho^2 - t^2) \geq h(R^2) > 0$, since h is decreasing and $\rho \leq |P| \leq R$. In particular, $t \geq h(R^2)/(2R)$ there, giving positive separation from the cone on each bounded portion of the cap.

The geometry lemma uses the outward cone normal, so its height is $-\eta$. With $\kappa = 5/2$ and $B = (5/2)^{3/8}/2$, its expansion gives, as $\rho \to \infty$,

$$\eta(\rho) = B\rho^{-\kappa} - \frac{15B}{16}\rho^{-\kappa-4} + O_5(\rho^{-\kappa-7}). \tag{3.2}$$

The notation $O_5(\rho^{-\kappa-7})$ means that the j-th derivative of the remainder is $O(\rho^{-\kappa-7-j})$ for $0 \leq j \leq 5$. In particular, at sufficiently large ρ we have $\eta(\rho) \asymp \rho^{-\kappa}$ and $|\eta^{(j)}(\rho)| \leq C_j\rho^{-\kappa-j}$ for $1 \leq j \leq 3$. The same lemma, using the strict-sign theorem of [29, Theorem 4.9], supplies

$$H_\sigma[E](p) \leq -c_\sigma(1 + \rho(p))^{-1-\sigma-\kappa}, \qquad p \in \partial E, \tag{3.3}$$

for a constant $c_\sigma > 0$, including at the axis. Because $\rho(p) \leq |p|$, the bound also holds with $|p|$ in place of $\rho(p)$. The source uses the same curvature sign and writes s for the perimeter order. In our notation that order is $\sigma = 2s \in (0, 1)$.

For $E_\lambda = \lambda E$, both ρ and t scale by λ, so $\eta_\lambda(\rho) = \lambda\eta(\rho/\lambda)$. For any measurable set G, whenever $H_\sigma[G](p)$ exists, one has

$$H_\sigma[\lambda G](\lambda p) = \lambda^{-\sigma} H_\sigma[G](p) \qquad (\lambda > 0). \tag{3.4}$$

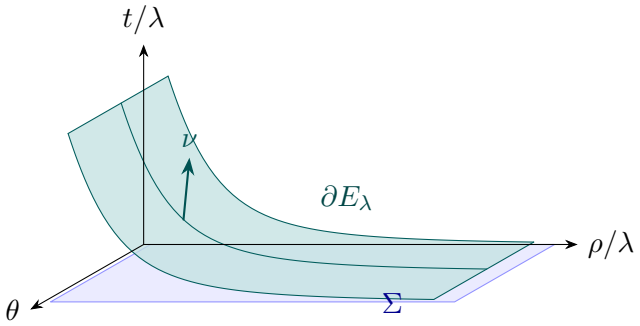


FIGURE 1. Projection of the cone $t = 0$ and the cap boundary $t = \eta_\lambda(\rho)$ in the scaled coordinates $(\rho/\lambda, t/\lambda)$, with θ a local angle in the y-block. The arrow represents the inward unit normal ν in the (ρ, t)-plane, pointing toward the side $t > \eta_\lambda(\rho)$.

Indeed $Q = \lambda Y$ multiplies the truncated integral by $\lambda^{-\sigma}$ and changes the deleted radius from ε to ε/λ.

We will also use a local estimate when G occupies one side of a smooth hypersurface near $p \in \partial G$, with inward unit normal ν. In a tubular neighborhood the signed distance, taken positive on G, is smooth. Its gradient at p is ν, since the distance vanishes along the interface and increases at unit speed on the inward normal. Taylor's formula gives $\nu \cdot h + O(|h|^2)$ at $p + h$. Thus the signs of G and its tangent half-space can differ only when $|\nu \cdot h| \le C|h|^2$.

For a uniform point on the unit sphere S^{n-1}, $n \ge 3$, the coordinate in any unit direction has density proportional to $(1-u^2)^{(n-3)/2}$ on $(-1, 1)$. Indeed the parametrization $(u, \sqrt{1-u^2}\,\omega)$, $\omega \in S^{n-2}$, scales the $n-2$ angular directions by $\sqrt{1-u^2}$ and its u-direction has length $1/\sqrt{1-u^2}$. The density is bounded. For $h = r\omega$ the discrepancy requires $|\nu \cdot \omega| \le Cr$, so its angular fraction is $O(r)$. For ε smaller than the tubular radius this gives

$$\int_{|h|<\varepsilon} \frac{\big|2\mathbf{1}_G(p+h) - 1 - \operatorname{sgn}(\nu \cdot h)\big|}{|h|^{8+\sigma}}\, dh \le C \int_0^\varepsilon r^{-\sigma}\, dr = O(\varepsilon^{1-\sigma}). \tag{3.5}$$

The estimate is uniform when the local signed-distance Hessian and the size of the neighborhood are controlled uniformly. The half-space sign has zero integral on centered annuli, so (3.5) also bounds the error made by truncating the principal value at ε. Figure 1 illustrates the choice of inward normal.

4. Estimates for the eight-dimensional saddle

Use the coordinates a, b, ρ, t of Section 3, and write $d_\Sigma = t$ and $T(x, y) = (y, x)$.

Lemma 4 (Differentiation and decay near the wells)**.** *For a bounded smooth solution of $Lv = f(v)$ in $\mathbb{R}^n$, each coordinate derivative satisfies $L(\partial_j v) = f'(v)\partial_j v$. If $n = 8$ and $v(X) \to \pm 1$ uniformly as $d_\Sigma(X) \to \pm\infty$, respectively, then*

$$\big|\, 1 - |v(X)| \,\big| + |\nabla v(X)| \le C(1 + |d_\Sigma(X)|)^{-\sigma} \qquad (X \in \mathbb{R}^8).$$

Proof. Since v is bounded and locally Lipschitz, its pointwise equation also holds in distributions by Lemma 2. The interior estimates on translated unit balls give uniform bounds on ∇v and $D^2 v$. Thus we may differentiate the equation using difference quotients. For a difference quotient of v in a coordinate direction, its increments at displacement k are bounded by $\|D^2 v\|_\infty |k|$ for $|k| \le 1$ and by $2\|\nabla v\|_\infty$ for $|k| > 1$. Both bounds are integrable against $K_n(k)$ on the indicated regions because $0 < \sigma < 1$. Subtract the equations at the two translated points and divide by the translation length. Dominated convergence on the kernel side and differentiability on the reaction side give $L(\partial_j v) = f'(v)\partial_j v$.

Fix X with $D_0 = d_\Sigma(X) > 0$ and let $R = D_0/2$. The signed distance is at least $D_0/2$ on $B_R(X)$, so the positive end condition gives $v \ge 3/4$ there for large D_0. On that ball,

$$(L + v(1+v))(1-v) = 0, \qquad (L + 3v^2 - 1)\partial_j v = 0.$$

Both zeroth-order coefficients are bounded below by $c_* = 11/16 > 0$. We compare each of $1 - v$, $\partial_j v$, and their negatives to the same barrier.

Choose a smooth function $0 \le \zeta \le 1$ which vanishes near zero and is one outside B_1. The kernel integral of $L\zeta$ is bounded by using the Lipschitz increment at distances below one and the bounded increment in the tail. Let M exceed the global bounds of the functions being compared. Set

$$B(P) = AR^{-\sigma} + M\zeta((P - X)/R), \qquad A \ge M\|L\zeta\|_\infty / c_*.$$

The barrier dominates each compared function outside $B_R(X)$. By scaling, $LB(P) = MR^{-\sigma}(L\zeta)((P - X)/R)$. For either coefficient $c(P) \ge c_*$, this gives

$$(L + c(P))B(P) \ge R^{-\sigma}\big(c_* A - M\|L\zeta\|_\infty\big) \ge 0.$$

If a compared function u exceeded B, then $u - B$ would attain a positive global maximum inside the ball. At that point $L(u - B) \ge 0$ by positivity of the kernel and $c(P)(u - B) > 0$, contrary to the equation and the barrier inequality. At the center X we obtain $|1 - v(X)| + |\nabla v(X)| \le C D_0^{-\sigma}$. Apply the same argument to $-v$ for large negative d_Σ, using oddness of f. On the remaining region of bounded $|d_\Sigma|$, the estimate follows from the global bounds on v and ∇v. □

Let S be the saddle of [12, Theorem 1.2(i)–(ii)] with both blocks of dimension four and with their exponent $\gamma = s$. Their trace operator is the fractional Laplacian with Fourier symbol $|\xi|^{2s}$, so its kernel is the isotropic K_8 of Section 2. The hypotheses hold for the smooth odd cubic, with $f(0) = f(1) = 0$ and $f'' < 0$ on $(0, 1)$. Their two radial variables (s, t)

correspond to (b, a) here, so $S > 0$ in D. The solution S is invariant under the two block rotations, odd under T, and takes values in $(-1, 1)$. On Σ the two radii are equal, so swapping them leaves the value of S unchanged. Oddness under the swap then gives $S = 0$ on all of Σ. Lemma 2 gives smoothness and uniform derivative bounds on translated unit balls. The profile in the cited asymptotics is the trace of their extension layer u_0. Their Dirichlet-to-Neumann identity preceding (1.3) identifies the trace equation in (1.5) with $Lu_0(\cdot, 0) = f(u_0(\cdot, 0))$ in one dimension. By (2.4), this one-dimensional operator is also the reduction of the eight-dimensional operator on a function of one linear coordinate. The trace has strictly positive derivative, end values ± 1, and value zero at the origin by (1.5). One-dimensional uniqueness from Section 2 identifies it with q. The asymptotics therefore give

$$S(X) - q(d_\Sigma(X)) \longrightarrow 0 \quad \text{uniformly as } |X| \to \infty,$$

and the same convergence for the spatial gradient at points off the factor axes. Since $|d_\Sigma(X)| \le |X|$, the convergence of the values and the limits of q give the uniform end conditions in Lemma 4.

Write $S = S(\rho, d)$ off the factor axes, with $d = d_\Sigma$. In D, the trace of [12, Proposition 1.8(i)–(ii)] gives $S_\rho \ge 0$ and $S_a < 0$ for $a > 0$, where S_a is the derivative in a at fixed b. The coordinate identities $\sqrt{2}S_a = S_\rho - S_d$ and $\sqrt{2}S_b = S_\rho + S_d$ give $S_d > 0$ and $S_b > 0$ there. Oddness under the block swap gives $S_a < 0 < S_b$ and $S_d > 0$ in the opposite chamber off the axes. These signs give weak block monotonicity throughout the quadrant by continuity, including on the cone and the axes. They also give strict increase of $S(\rho, d)$ in d on $(-\rho, \rho)$, by integration on the intervals on each side of $d = 0$.

Lemma 5 (Dilation derivative of the saddle)**.** *The function $P(X) = X \cdot \nabla S(X)$ and its first derivatives are bounded in $\mathbb{R}^8$. In particular, off the factor axes,*

$$|S_\rho(\rho, d)| \le \frac{C}{\rho} \qquad (|d| < \rho),$$

and for each $M > 0$,

$$|S_{\rho d}(\rho, d)| \le \frac{C_M}{\rho} \qquad (|d| \le M,\ \rho \ge 2M + 1).$$

Proof. We will compare $S(\lambda X)$ to $S(X)$ for λ just above one. The comparison uses a positive function for the linearized equation on the odd class. We first derive the angular kernel bounds that give its strict sign, including at small d.

Angular averaging. Let $0 \le a < b$ denote the block radii of $X \in D$, and $0 \le c < e$ those of an integration point in D. Use probability measure $d\bar{\omega}$ on each copy of S^3 and radial measure $d\mu = |S^3|^2 c^3 e^3 \, dc \, de$. Let K_{d} be the angular average of $c_{8,s}|X - Y|^{-8-\sigma}$, and let K_{c} be the same average with the two integration radii interchanged. Set $J = K_{\mathrm{d}} - K_{\mathrm{c}}$. For an invariant

function u odd under T, split its fractional integral into the two chambers and change variables by T on the second. On D this gives

$$\begin{aligned} L_{\mathrm{o}}u(X) &= \int_{c<e} (u(X)-u(c,e))J(a,b;c,e)\,d\mu + A(X)u(X),\\ A(X) &= 2\int_{c<e} K_{\mathrm{c}}(a,b;c,e)\,d\mu. \end{aligned} \tag{4.1}$$

To see the multiplication term, the integral over the opposite chamber becomes $(u(X)+u(c,e))K_{\mathrm{c}}$ after the swap. Adding the direct term $(u(X)-u(c,e))K_{\mathrm{d}}$ gives the displayed formula. In Cartesian coordinates, $A(X) = 2\int_{D^c} K_8(X-Y)\,dY$, which is finite and positive since X is at positive distance from D^c. The formula applies to bounded invariant odd functions that are locally Lipschitz in D. A Lipschitz increment gives absolute integrability at the evaluation point, and the part in the opposite chamber has no singularity there.

Set $B(z) = \int_{S^3} e^{z\omega_1}\,d\bar{\omega}$ and $p = (8+\sigma)/2$. By

$$w^{-p} = \frac{1}{\Gamma(p)}\int_0^\infty e^{-\tau w}\tau^{p-1}\,d\tau,$$

the Gaussian factors for the direct and cross kernels at $\tau > 0$ are

$$\begin{aligned} M_{\mathrm{d}} &= e^{-\tau(a^2+b^2+c^2+e^2)}B(2\tau ac)B(2\tau be),\\ M_{\mathrm{c}} &= e^{-\tau(a^2+b^2+c^2+e^2)}B(2\tau ae)B(2\tau bc). \end{aligned}$$

Their logarithmic ratio is

$$\log\frac{M_{\mathrm{d}}}{M_{\mathrm{c}}} = \int_a^b\int_c^e 2\tau\big[(\log B)'(2\tau uv) + 2\tau uv\,(\log B)''(2\tau uv)\big]\,dv\,du. \tag{4.2}$$

Indeed the logarithm of the ratio is the rectangular difference of the function $(u,v)\mapsto \log B(2\tau uv)$, so the fundamental theorem of calculus integrates its mixed derivative. Here $(\log B)'(0) = 0$ by reflection on the sphere, and $(\log B)''$ is the strictly positive variance of ω_1 under the probability measure proportional to $e^{z\omega_1}d\bar{\omega}$. Thus the integrand in (4.2) is nonnegative. The Gaussian representation gives, off the orbit diagonal,

$$J(a,b;c,e) = \frac{c_{8,s}}{\Gamma(p)}\int_0^\infty (M_{\mathrm{d}}-M_{\mathrm{c}})\tau^{p-1}\,d\tau \geq 0.$$

Both kernel averages are finite there. The orbit diagonal has $d\mu$-measure zero.

We need two lower bounds, both obtained by restricting the Gaussian integral to $1 \leq \tau \leq 2$. If a, b are bounded and (c,e) runs in a compact rectangle contained in $0 < c < e$, the variance has a positive lower bound on the bounded range of $2\tau uv$. Integrating $(\log B)''$ from zero gives $(\log B)'(2\tau uv) \geq c_1 uv$. The u, v integrals of uv give $(b^2-a^2)(e^2-c^2)/4$, with e^2-c^2 bounded below on the rectangle. Thus the logarithmic ratio is at least $c_2(b^2-a^2)$. Since M_{c} has a positive minimum on these compact

parameter sets, writing $\ell = \log(M_{\mathrm{d}}/M_{\mathrm{c}})$ gives $M_{\mathrm{d}} - M_{\mathrm{c}} = M_{\mathrm{c}}(e^{\ell} - 1) \geq M_{\mathrm{c}}\ell$. Integration over $1 \leq \tau \leq 2$ yields

$$J(a, b; c, e) \geq c_3(b^2 - a^2). \tag{4.3}$$

For the second bound let $\rho = (a+b)/\sqrt{2}$ be large and $0 < d = (b-a)/\sqrt{2} < 1$. Integrate over the strip

$$|\rho' - \rho| \leq 1, \qquad 1 \leq d' \leq 2, \qquad \rho' = (c+e)/\sqrt{2}, \quad d' = (e-c)/\sqrt{2}.$$

All four radii are comparable to ρ. The squared cross distance in radial variables is

$$(a-e)^2 + (b-c)^2 = (\rho - \rho')^2 + (d + d')^2 \leq 10.$$

In the rectangular integral (4.2), $2\tau uv \geq 1$ for large ρ. Since $(\log B)'$ increases and is positive at 1, the integrand has a positive lower bound. The side lengths $b-a = \sqrt{2}d$ and $e-c = \sqrt{2}d' \geq \sqrt{2}$ thus give a lower bound $c_4 d$ for the logarithmic ratio. For $z \geq 1$, restrict the spherical integral defining $B(z)$ to $1 - z^{-1} \leq \omega_1 \leq 1$, where $e^{z\omega_1} \geq e^{z-1}$. The coordinate density on S^3 is proportional to $(1 - \omega_1^2)^{1/2}$. On this interval it is bounded below by a constant times $(1 - \omega_1)^{1/2}$, whose integral is a constant times $z^{-3/2}$. It follows that $B(z) \geq c_5 e^z z^{-3/2}$. Using this bound in the two factors of M_{c} gives

$$M_{\mathrm{c}} \geq c_6 \rho^{-6} e^{-\tau((a-e)^2 + (b-c)^2)} \geq c_7 \rho^{-6} \qquad (1 \leq \tau \leq 2).$$

Hence $J \geq c_8 d\, \rho^{-6}$ on the strip. Since $c^3 e^3 \asymp \rho^6$ and the change of variables $(c, e) \mapsto (\rho', d')$ is orthogonal, integration gives

$$\int_{|\rho' - \rho| \leq 1,\ 1 \leq d' \leq 2} J(a, b; c, e)\, d\mu \geq c_9 d. \tag{4.4}$$

For every $X \in D$, the multiplication coefficient satisfies

$$A(X) \geq c_{10} d^{-\sigma}. \tag{4.5}$$

Indeed, keep the block directions and swap the radii of X. If one block vanishes, choose its direction arbitrarily. The resulting point $X^\sharp$ has signed distance $-d$ and is $2d$ from X. The ball $B_{d/2}(X^\sharp)$ lies in D^c by the Lipschitz bound for d_Σ. On this ball $|X - Y| \leq 5d/2$, so integrating $K_8(X - Y)$ there gives at least $c d^8 d^{-8-\sigma}$.

Comparison on fixed balls. For every $R > 0$ we claim

$$S(X) \geq c_R(b^2 - a^2) \qquad (X \in D \cap B_R). \tag{4.6}$$

Let $\phi(X) = (|y|^2 - |x|^2)\chi(|X|^2)$, with χ smooth and nonnegative, equal to one on $[0, R^2]$ and vanishing for $r \geq (R+1)^2$. Choose a smooth bump ψ invariant under both block rotations, nonnegative on D and supported on a compact subset of $D \setminus \overline{B}_{R+2}$ away from the factor axes. Take it positive on a

closed rectangle $\mathcal{Q}$ of orbit radii $0 < c < e$ with nonempty interior. Extend it by oddness. At $X \in B_{R+1} \cap D$, $\psi(X) = 0$, so (4.1) and (4.3) give

$$L_{\mathrm{o}}\psi(X) = -\int_{c<e} \psi(Y) J(X,Y)\, d\mu(Y) \le -c(b^2 - a^2) \int_{\mathcal{Q}} \psi(Y)\, d\mu(Y).$$

The last integral is strictly positive. Also $L\phi$ is smooth, invariant and odd, by differentiation under the absolutely convergent increment integral. For any smooth invariant odd G one has

$$|G(ae_1, be_1)| \le C_R(b^2 - a^2), \qquad 0 \le a \le b \le R+1.$$

In fact, $G(ae_1, ae_1) = 0$, and the derivative in the second radius vanishes at zero by evenness. Its magnitude on the compact square is bounded by a constant times the second radius, using the second derivative bound. Integration from a to b gives the claim.

Apply this bound to $L\phi$ and choose M_0 large enough that $L_{\mathrm{o}}(\phi + M_0\psi) \le 0$ on $B_{R+1} \cap D$. For $\varepsilon > 0$ small, $w = \varepsilon(\phi + M_0\psi)$ lies below S outside that region in D, by strict positivity on the compact support of ψ there. Both functions vanish at the cone. If $w - S$ were positive anywhere in D, it would attain its positive maximum on $\overline{D} \cap \overline{B}_{R+1}$, and the boundary inequalities would put the maximum in $B_{R+1} \cap D$. The nonnegative J and positive A would imply $L_{\mathrm{o}}(w - S) > 0$ at the maximum, while $Lw \le 0 \le f(S) = LS$ there. This proves (4.6).

The dilation derivative $P(X) = X \cdot \nabla S(X)$ is smooth, block invariant and odd, since S has these properties and dilations commute with the block symmetries. In the next estimate the second derivatives of P on a compact ball are controlled by the derivatives of S through order three, supplied by Lemma 2. The same bound for $G = P$ on B_{2R} gives $|P(\ell X)| \le C_{2R}\ell^2(b^2 - a^2)$ when $X \in B_R \cap D$ and $1 \le \ell \le 2$. Integrate $\frac{d}{d\ell} S(\ell X) = \ell^{-1} P(\ell X)$ from 1 to $\lambda \le 2$, then use (4.6) to obtain

$$S(\lambda X) - S(X) \le C_R(\lambda - 1) S(X) \qquad (X \in D \cap B_R). \tag{4.7}$$

A strict linearized inequality outside a ball. Fix $0 < \vartheta < 1$ and set $F(u) = u - \vartheta u^3$. On D, $(1 - \vartheta)S \le F(S) \le S$. For $u, v \ge 0$,

$$F(u) - F(v) - F'(u)(u - v) = \vartheta(v - u)^2(v + 2u).$$

Subtract $F'(S(X)) L_{\mathrm{o}} S(X)$ from $L_{\mathrm{o}}(F(S))(X)$ in (4.1). The integral part becomes the displayed cubic remainder with $u = S(X)$, $v = S(Y)$. The multiplication part becomes $A(X)(F(S) - F'(S)S) = 2\vartheta A(X) S^3$, with $S = S(X)$. Since $L_{\mathrm{o}} S = f(S)$ and $F'(S) f(S) - f'(S) F(S) = 2(1 - \vartheta) S^3$, we obtain

$$\begin{aligned} (L_{\mathrm{o}} - f'(S)) F(S)(X) &= 2(1 - \vartheta) S(X)^3 + 2\vartheta A(X) S(X)^3 \\ &\quad + \vartheta \int_{c<e} (S(Y) - S(X))^2 \qquad (4.8) \\ &\quad \times (S(Y) + 2S(X)) J(X,Y)\, d\mu(Y). \qquad (4.9) \end{aligned}$$

For large ρ, the saddle asymptotics give $S(Y) \geq q(1)/2 =: m_0 > 0$ on the strip in (4.4). Also $S(X) \leq Cd$, since S vanishes on Σ and has bounded gradient. Choose $d_0 \in (0,1)$ small enough that $S(X) \leq m_0/2$ for $d \leq d_0$. Then on the strip the product $(S(Y) - S(X))^2(S(Y) + 2S(X))$ is at least $m_0^3/4$. By (4.4), the integral in (4.9) is bounded below by a positive constant times d, and hence by a positive constant times $S(X)$. When $d \geq d_0$, the saddle asymptotics and $q(d_0) > 0$ give $S(X) \geq m_1 > 0$ outside a sufficiently large ball. The first term in (4.9) supplies the desired bound in this range. Thus there are $c_0 > 0$ and $R_0 < \infty$ with

$$(L_{\mathrm{o}} - f'(S))F(S) \geq c_0 S \qquad (X \in D \setminus B_{R_0}). \tag{4.10}$$

The dilation comparison. Put $S_\lambda(X) = S(\lambda X)$ and $\delta = \lambda - 1 > 0$. The scaled equation is $LS_\lambda = \lambda^\sigma f(S_\lambda)$. Let $V = S + C\delta F(S)$. With $C\delta \leq 1$, we have $f(V) \leq V \leq 2S$ on D. Concavity of f on $[0,\infty)$ and $\lambda^\sigma - 1 \leq \sigma\delta$ give, by (4.10),

$$\begin{aligned} L_{\mathrm{o}}V - \lambda^\sigma f(V) &\geq C\delta(L_{\mathrm{o}} - f'(S))F(S) - (\lambda^\sigma - 1)f(V) \\ &\geq \delta(Cc_0 - 2\sigma)S \qquad (X \in D \setminus B_{R_0}). \end{aligned}$$

Choose C large enough for this to be positive and for $S_\lambda \leq V$ on $D \cap B_{R_0}$, using (4.7) and $F(S) \geq (1 - \vartheta)S$.

The comparison at infinity uses the relative layer asymptotics

$$\sup_{\substack{X \in D \\ |X| > R}} \left| \frac{S(X)}{q(d_\Sigma(X))} - 1 \right| \longrightarrow 0 \qquad (R \to \infty). \tag{4.11}$$

For $d \geq 1$, this follows from the uniform value asymptotics and $q(d) \geq q(1) > 0$. For $0 < d \leq 1$ and large $|X|$, join the cone to X by keeping ρ and both block directions fixed and varying the signed distance from 0 to d. This is a segment of length d, with radii $(\rho - r)/\sqrt{2}$ and $(\rho + r)/\sqrt{2}$ at distance r. It stays off the factor axes and at distance at least $\rho \geq |X|/\sqrt{2}$ from the origin. At the cone endpoint $S = 0$ and $q(d_\Sigma) = q(0) = 0$ by the centering in Section 2. Integrating their gradient difference bounds the value difference by d times a quantity tending uniformly to zero. On $[0,1]$, the continuous positive derivative q' has a positive minimum. Thus $q(d) \geq cd$ there, which gives (4.11) also for $0 < d \leq 1$.

For $z \geq 1$, the layer derivative estimate in [3, Theorem 2.7] gives $q'(z) \leq Cz^{-1-\sigma}$. Its hypotheses hold by the regularity and well conditions checked in Section 2, in particular $f'(\pm 1) = -2 < 0$. Using $q(z) \geq q(1)$ for $z \geq 1$ and $q(z) \geq cz$ for $0 < z \leq 1$, we obtain $zq'(z)/q(z) \leq C_q$ for all $z > 0$. Integrating the logarithmic derivative from d to λd gives $q(\lambda d)/q(d) \leq \lambda^{C_q}$. For $1 \leq \lambda \leq 2$, the mean value theorem bounds this ratio above by $1 + C_q'\delta$.

For each $0 < \delta \leq 1$, choose an outer comparison radius greater than R_0 so that the relative error in (4.11) is at most $\delta/4$ outside that radius. If X

is outside, so is λX, whose cone distance is λd. Therefore

$$\frac{S_\lambda(X)}{S(X)} \le \frac{1+\delta/4}{1-\delta/4}\frac{q(\lambda d)}{q(d)} \le 1 + C_q''\delta,$$

where C_q'' is independent of δ and the radius. Choose $C(1-\vartheta) \ge C_q''$, in addition to the earlier conditions on C, and then restrict δ by $C\delta \le 1$. This gives $S_\lambda \le V$ on the outer region as well as on the inner ball.

To compare in the remaining annulus write $h = S_\lambda - V$ and $\mathcal{A}_\lambda = L_{\mathrm{o}} - \lambda^\sigma f'(S)$. At a point where $h > 0$ in the exterior of B_{R_0}, concavity and $V \ge S$ imply

$$f(S_\lambda) - f(V) \le f'(V)h \le f'(S)h, \qquad \mathcal{A}_\lambda h \le 0.$$

Since $f'(S) \le 1$ and $0 \le F(S) \le S$ in D, (4.10) gives

$$\mathcal{A}_\lambda F(S) \ge \big(c_0 - (\lambda^\sigma - 1)\big)S \ge \tfrac{c_0}{2}S \qquad \text{in } D \setminus B_{R_0}$$

for sufficiently small δ. Use the positive divisor $\Psi = F(S) + \eta$ on D, extended oddly, for a constant $\eta > 0$ to be chosen. Its jump at the cone is a positive distance from each evaluation point in D, so the operator there is well defined. The constant η on D drops out of the first term of (4.1) and contributes $\eta(A - \lambda^\sigma f'(S))$ to $\mathcal{A}_\lambda\Psi$. Choose $d_* \in (0,1)$ so that $c_{10}d_*^{-\sigma} \ge 2$. For $d \le d_*$, (4.5) and $\lambda^\sigma \le 2$ imply that this contribution is nonnegative. For $d \ge d_*$, the uniform saddle asymptotics give a positive lower bound for S outside a large ball, and positivity on the remaining compact set with $d \ge d_*$ extends the bound. Thus the bound holds with the original R_0 used to choose C. Write $m_* > 0$ for a common lower bound. Since $A - \lambda^\sigma f'(S) \ge -2$, choosing $0 < \eta < c_0 m_*/8$ gives $\mathcal{A}_\lambda\Psi > 0$ throughout the exterior. If h were positive somewhere, h/Ψ would attain a positive maximum m inside $D \setminus B_{R_0}$. Indeed, the inner and outer comparisons make the quotient nonpositive at the radial boundaries of a bounded annulus. On the cone portion of its boundary the quotient extends continuously by zero because $h = 0$ there and $\Psi \ge \eta$ on D. A positive maximum therefore occurs at an interior point X_* in D outside the inner ball. The function $h - m\Psi$ is nonpositive on D and vanishes at X_*. At this point the multiplication terms in $\mathcal{A}_\lambda(h - m\Psi)$ vanish because $(h - m\Psi)(X_*) = 0$. Thus

$$\mathcal{A}_\lambda(h - m\Psi)(X_*) = -\int_{c<e} (h - m\Psi)(Y)J(X_*, Y)\, d\mu(Y) \ge 0.$$

Since $m > 0$ and $\mathcal{A}_\lambda\Psi(X_*) > 0$, this implies $\mathcal{A}_\lambda h(X_*) > 0$, a contradiction.

Off the axes in D the radial monotonicities give $P = \rho S_\rho + d S_d \ge 0$, and continuity extends this to the axes. Integration along rays thus gives $S_\lambda \ge S$ in D. The upper comparison just proved bounds $(S_\lambda - S)/\delta$ on D for all sufficiently small $\delta > 0$. Oddness gives a global absolute bound. Letting $\delta \downarrow 0$ gives $P \in L^\infty(\mathbb{R}^8)$. On D it follows that $\rho S_\rho \le P$, and oddness and continuity extend the bound for $|S_\rho|$ to $|d| < \rho$.

It remains to bound ∇P. Take the difference quotient of the scaled equation and pass to the limit in distributions. For a compactly supported

smooth test function φ, $L\varphi$ is bounded and decays as $O(|X|^{-8-\sigma})$ outside a large ball, so the global quotient bound permits dominated convergence against $L\varphi$. On the source side write

$$\frac{\lambda^\sigma f(S_\lambda) - f(S)}{\delta} = \lambda^\sigma \frac{f(S_\lambda) - f(S)}{\delta} + \frac{\lambda^\sigma - 1}{\delta} f(S).$$

Both terms are uniformly bounded by the quotient estimate and smoothness of the cubic on $[-1, 1]$. This proves $LP = f'(S)P + \sigma f(S)$ in distributions. Use the bounded-source and interior Schauder estimates of [17, Theorem 1.1(a) and Corollary 3.5]. Write the even kernel of $-L$ in polar coordinates. The radial factor is $r^{-1-\sigma}\,dr$ and the angular factor is a positive constant times surface measure on S^7. The resulting finite spectral measure has positive directional ellipticity integral independent of the direction. The constant density is Hölder, which permits the local Schauder estimate with only a global L^∞ bound on P. The source is bounded, so P first belongs to C^σ on smaller balls. The coefficients $f'(S)$ and $f(S)$ are bounded and Lipschitz by the derivative bounds for S. Whenever P is C^a for $0 < a < 1$, the source is therefore C^a, and the Schauder gain gives $C^{a+\sigma}$ on a smaller ball when $a+\sigma$ is noninteger. Choose $0 < b < \sigma$ so that jb and $jb+\sigma$ are noninteger for all positive integers j. For each j there are at most countably many forbidden values, so their union cannot exhaust the interval. Start with C^b. As long as $jb < 1$, the gain from C^{jb} to $C^{jb+\sigma}$ gives at least $C^{(j+1)b}$ on a smaller ball. Stop at the first step where $jb + \sigma > 1$, which occurs in finitely many steps with $jb < 1$. One can use a fixed finite sequence of nested radii between one and one half on every translated unit ball. Thus the estimates are uniform in the center, and ∇P is bounded globally. The global bound on $D^2 S$ follows directly from Lemma 2.

Finally, at fixed ρ and fixed block directions ω, ω', varying d traces the affine line

$$X = \left(\frac{\rho - d}{\sqrt{2}}\omega, \frac{\rho + d}{\sqrt{2}}\omega' \right).$$

It has unit velocity, so the derivatives in d are ordinary Cartesian directional derivatives on the region with both radii positive. Differentiating $P = \rho S_\rho + dS_d$ gives $\rho S_{\rho d} = P_d - S_d - dS_{dd}$. The global derivative bounds control the right-hand side on each fixed band $|d| \le M$. □

5. Transport in the exterior of the cap

We keep the cone coordinates of Section 3, with $a = |x|$ and $b = |y|$. Near the cap the map sends signed normal distance to the cone coordinate t, and arclength along the cap's meridian to the cone coordinate ρ. At the cap its differential is therefore an isometry in the radial plane. In the six perpendicular directions it dilates the two angular blocks. The corresponding change in the affine kernel has zero integral in those six directions for each fixed radial displacement. We use this cancellation after restricting the profile to the radial plane.

Proposition 6. *Use the exponent and operator conventions of Section 2 and the curvature convention of Section 3. Let v be a smooth solution of $Lv = f(v)$ on $\mathbb{R}^8$ with values in $(-1,1)$. Assume that v depends only on a, b, is nonincreasing in a and nondecreasing in b, and converges uniformly to the corresponding well as $t \to \pm\infty$.*

Let $E_1 \subset D$ be a smooth cap invariant under the block rotations. Suppose that outside a compact set membership in E_1 is equivalent to $t > \eta_1(\rho)$, where $\eta_1 > 0$ is smooth and, for some $0 < \kappa < 7$, satisfies

$$\eta_1(q) \asymp q^{-\kappa}, \qquad |\eta_1^{(j)}(q)| \le C q^{-\kappa-j} \quad (1 \le j \le 3).$$

For $\lambda \ge 1$ put $E_\lambda = \lambda E_1$, $\eta_\lambda(q) = \lambda \eta_1(q/\lambda)$, and

$$m_\lambda(r) = \lambda^{\kappa+1} r^{-\kappa-1-\sigma}.$$

There exist $L_0 > 1$ and smooth, uniformly bi-Lipschitz diffeomorphisms $\Phi_\lambda = \lambda \Phi_1(\cdot/\lambda)$ commuting with the block rotations, with the following properties for every fixed $L_1 \ge L_0$.

(1) *The function $u_\lambda = v \circ \Phi_\lambda$ satisfies $u_\lambda \le v$. The sets $F_\lambda = \Phi_\lambda^{-1}(D)$ and E_λ coincide outside $B_{L_1\lambda}$. There is $c > 0$ such that at $P = p + d\nu(p)$, with $p \in \partial E_\lambda$, ν pointing into the cap, $|P| \ge L_1\lambda$, and $|d| \le c|p|$, the map preserves the two block directions and has radial coordinates*

$$(\rho, t)\big(\Phi_\lambda(P)\big) = \big(S_\lambda(\rho(p)), d\big), \qquad S_\lambda(q) = q - \int_q^\infty \Big(\sqrt{1 + \eta_\lambda'(u)^2} - 1\Big)\, du.$$

The arclength label satisfies $S_\lambda(q) \asymp q$ uniformly for $q \ge L_0\lambda/\sqrt{2}$ after increasing L_0 if needed. On the tube Φ_λ strictly increases a and strictly decreases b.

(2) *If $V_\lambda : \mathbb{R}^8 \to [-1,1]$ is measurable and equals u_λ outside $B_{L_1\lambda}$, then*

$$|(LV_\lambda - f(V_\lambda))(P)| \le C_{L_1} m_\lambda(|P|), \qquad |P| \ge 2L_1\lambda.$$

(3) *Suppose also that*

$$\lambda^{-8} \int_{B_{L_1\lambda}} |V_\lambda - (2\chi_{E_\lambda} - 1)|\, dQ \longrightarrow 0.$$

For every finite T, as $\lambda \to \infty$,

$$(LV_\lambda - f(V_\lambda))(P) = c_{8,s} H_\sigma[E_\lambda](p) + o\big(m_\lambda(|P|)\big) \tag{5.1}$$

uniformly at $P = p + d\nu(p)$ with $|d| \le T$ and $|P| \ge 2L_1\lambda$. In particular, if $H_\sigma[E_1](p) \le -c(1+|p|)^{-\kappa-1-\sigma}$ on ∂E_1, the residual in (5.1) is at most $-c_* m_\lambda(|P|)$ for a constant $c_* > 0$ independent of T once λ is sufficiently large depending on T.

Proof. The end-state hypothesis gives agreement of the signs of v and t for large $|t|$. Thus Lemma 4, with the constant enlarged for bounded $|t|$, gives

$$|v(X) - (2\chi_D(X) - 1)| + |\nabla v(X)| \le C(1 + |t(X)|)^{-\sigma}. \tag{5.2}$$

The map. We work first in the plane of block radii. Put

$$e_\lambda(R) = \lambda^{\kappa+1}(\lambda + R)^{-\kappa}, \qquad \delta_\lambda(R) = \frac{e_\lambda(R)}{\lambda + R}.$$

The height assumptions and dilation give $\eta_\lambda(q) \asymp e_\lambda(q)$ and $|\eta_\lambda^{(j)}(q)| \le Ce_\lambda(q)q^{-j}$ for $1 \le j \le 3$ when q is beyond a fixed multiple of λ. These estimates hold with $e_\lambda(\rho)$ in place of $e_\lambda(q)$ whenever $\rho/2 \le q \le 2\rho$. The ratio $e_\lambda(\rho)/\rho$ tends uniformly to zero as ρ/λ grows.

Choose $c_0 > 0$ small and then $C_0 > 1$ large. In $\rho > C_0\lambda$, $|t| < c_0\rho$, the inward normal coordinates, with q the radial label at the footpoint, have the form

$$(\rho, t) = (q, \eta_\lambda(q)) + \frac{d}{\sqrt{1 + \eta_\lambda'(q)^2}}\,(-\eta_\lambda'(q), 1).$$

Eliminating d gives

$$\rho = q - (t - \eta_\lambda(q))\eta_\lambda'(q).$$

For $\rho/2 \le q \le 2\rho$ the right side is $q + O(e_\lambda(\rho))$, so it brackets ρ at the endpoints. Its derivative in q is $1+\eta_\lambda'(q)^2-(t-\eta_\lambda(q))\eta_\lambda''(q)$, which is within $1/2$ of one by the height bounds. Thus q exists uniquely and is smooth in (ρ, t).

A cap point at radial label ρ with the same block directions has distance at most $|t - \eta_\lambda(\rho)| < \rho/4$, by the choices of c_0, C_0. Any boundary point with radial label outside $[\rho/2, 2\rho]$ is at distance at least $\rho/2$, since ρ is 1-Lipschitz. At any fixed radii, choosing the same block directions minimizes the distance by the calculation in Section 3. The nearest point thus lies on the smooth graph in the interval, where minimizing distance requires the normal equation. Its normal parameter is d above. The sign agrees with $t - \eta_\lambda(\rho)$ because

$$t - \eta_\lambda(\rho) = \frac{d}{\sqrt{1 + \eta_\lambda'(q)^2}} + \eta_\lambda(q) - \eta_\lambda(\rho).$$

Since $|q - \rho| \le |d||\eta_\lambda'(q)|$, the mean value theorem gives

$$|\eta_\lambda(q) - \eta_\lambda(\rho)| \le |d| \sup_{[\rho/2, 2\rho]} |\eta_\lambda'|^2.$$

For C_0 large this is smaller than the magnitude of the first term whenever $d \ne 0$. This proves that d is the signed distance, positive in the cap.

The exact map sends (ρ, t) to $(S_\lambda(q), d)$. We first check its displacement estimates in the meridional plane. Write $F(q, t) = (t - \eta_\lambda(q))\eta_\lambda'(q)$, so $q = \rho + F(q, t)$. In this region $|F| \le Ce_\lambda(\rho)$, $|DF| \le Ce_\lambda(\rho)/\rho$, and $|D^2F| \le Ce_\lambda(\rho)/\rho^2$, with the derivatives taken in q, t. For example,

$$F_q = (t - \eta_\lambda)\eta_\lambda'' - (\eta_\lambda')^2, \qquad F_{qq} = (t - \eta_\lambda)\eta_\lambda^{(3)} - 3\eta_\lambda'\eta_\lambda''.$$

Implicit differentiation first bounds Dq, since $1 - F_q \ge 1/2$. Differentiating $q - \rho = F(q, t)$ then gives $|D(q-\rho)| \le Ce_\lambda(\rho)/\rho$ and $|D^2q| \le Ce_\lambda(\rho)/\rho^2$. For $S_\lambda(q) - q$, integrating the bound $\sqrt{1 + (\eta_\lambda')^2} - 1 \le C(\eta_\lambda')^2$ from q to infinity

gives $O(e_\lambda(q)^2/q)$. In particular, for q/λ large enough, $0 \le q - S_\lambda(q) \le q/2$, giving the stated comparability. The first and second derivatives of $S_\lambda(q)-q$ in q are $O(e_\lambda(q)^2/q^2)$ and $O(e_\lambda(q)^2/q^3)$, by its defining integral. For the other component, write

$$d - t = -\eta_\lambda(q) + (t - \eta_\lambda(q))\big(\sqrt{1+\eta_\lambda'(q)^2} - 1\big).$$

The height derivatives through order three give

$$\left|\partial_q^j\big(\sqrt{1+\eta_\lambda'(q)^2} - 1\big)\right| \le Ce_\lambda(q)^2 q^{-2-j}, \qquad j = 0,1,2.$$

Since $|t - \eta_\lambda(q)| \le C\rho$, the product rule and the bounds on Dq, D^2q give

$$|D^j(d-t)| \le Ce_\lambda(\rho)\rho^{-j}, \qquad j = 0,1,2.$$

The bounds on $q - \rho$ and $S_\lambda(q) - q$ give the same orders for $S_\lambda(q) - \rho$.

To extend the map, fix $0 < \tau < c_0/4$. Multiply the displacement in (ρ, t) by a product of smooth cutoffs with values in $[0,1]$. Use a radial cutoff supported in $\rho > C_0\lambda$ and equal to one for $\rho \ge 2C_0\lambda$, and an angular cutoff supported in $|t| < c_0\rho$ and equal to one for $|t| \le 2\tau\rho$. The cutoffs depend only on $\rho/(C_0\lambda)$ and t/ρ, respectively. Lift the resulting map by retaining both block directions. On its support $a, b \asymp \rho$, and the target radii are positive since the displacement is $o(\rho)$ as ρ/λ grows. The map is the identity off the support, including near either axis. Each derivative of a cutoff costs one inverse power of ρ. If the first radius changes by Δa, the first Cartesian block changes by $\Delta a\, x/|x|$. Since $|x| \asymp \rho$ on the support, the first two derivatives of the direction have bounds $C\rho^{-1}$ and $C\rho^{-2}$, and similarly for the second block. The cutoffs vanish smoothly at the boundary of the region where the normal coordinates were constructed. The Cartesian displacement and its first two derivatives therefore obey

$$\begin{aligned} |\Phi_\lambda(Q) - Q| &\le Ce_\lambda(|Q|), \\ |D\Phi_\lambda(Q) - I| &\le C\delta_\lambda(|Q|), \\ |D^2\Phi_\lambda(Q)| &\le Ce_\lambda(|Q|)(\lambda + |Q|)^{-2}. \end{aligned} \tag{5.3}$$

Increase C_0 with c_0, τ fixed so that the displacement has Lipschitz constant less than $1/4$. For each prescribed target Y, the map $Q \mapsto Y - (\Phi_\lambda(Q) - Q)$ is a contraction on $\mathbb{R}^8$, whose fixed point is the unique preimage of Y. The Lipschitz bound also gives the uniform inverse Lipschitz bound, and the inverse function theorem gives smoothness. The arclength integral gives $S_\lambda(q) = \lambda S_1(q/\lambda)$ by substitution. Together with the scaling of the height, normal coordinates and cutoffs this yields $\Phi_\lambda = \lambda\Phi_1(\cdot/\lambda)$.

Before cutting off, the change of t is $-\eta_\lambda(q) + d(1 - 1/\sqrt{1+\eta_\lambda'(q)^2})$. Here $|d| \le C(c_0 q + e_\lambda(q))$ and $|\eta_\lambda'| \le Ce_\lambda(q)/q$. Thus the second term is at most $\eta_\lambda(q)/4$ in absolute value if C_0 is large. The change of ρ is at most $|S_\lambda(q) - q| + |d\eta_\lambda'(q)|$ in absolute value. Dividing this by $\eta_\lambda(q) \asymp e_\lambda(q)$ bounds the ratio by $C(c_0 + e_\lambda(q)/q)$, which is at most $1/2$ for c_0 small and C_0 large. The change of t is therefore negative and dominates the change

of ρ. As $a = (\rho - t)/\sqrt{2}$ and $b = (\rho + t)/\sqrt{2}$, the exact map increases a and decreases b. The cutoffs preserve the corresponding weak inequalities, which give $u_\lambda \le v$.

Choose L_0 large enough that at $|P| \ge L_0\lambda$ the radial cutoff is one, using $\rho(P) \ge |P|/\sqrt{2}$. In the angular region $|t| \le 2\tau\rho$, the sign after transport is the sign of d, hence the cap sign. In its complement, $|t| > 2\tau\rho$. Choose L_0 large enough that $\eta_\lambda(\rho)$ and the change in t are at most $\tau\rho/4$ in absolute value throughout the exterior. The cap sign and transported cone sign then both agree with the original cone sign. This proves $F_\lambda = E_\lambda$ on the exterior.

For the tube assertion, if $|P| \ge L_1\lambda$ and $P = p + d\nu(p)$ with $|d| \le c|p|$, then $|p| \ge |P|/(1+c)$. Thus p belongs to the exterior graph for L_0 large. Both block radii there are comparable to $|p|$, and a short normal segment stays in the same radial plane without reaching either axis. Since t, ρ are 1-Lipschitz, $t(P) = O(e_\lambda(|p|) + c|p|)$ and $|\rho(P) - \rho(p)| \le c|p|$. Since $\rho(p) \ge |p|/\sqrt{2}$, we may choose c small enough that $\rho(p)$ lies in $[\rho(P)/2, 2\rho(P)]$. Also choose c small and L_0 large enough that $|t(P)| \le 2\tau\rho(P)$, where both cutoffs are one. The given normal segment satisfies the planar normal equation with $q = \rho(p)$. Uniqueness in this interval identifies p as the footpoint used by the exact transport.

Kernel estimates. In the calculations below suppress the index λ, put $K(h) = |h|^{-8-\sigma}$, and write $J(Q) = \det D\Phi(Q)$. The determinant $J(Q)$ is positive because $\|D\Phi - I\| < 1/4$. Use $f(u(P)) = (Lv)(\Phi(P))$ and change variables $Y = \Phi(Q)$ in the latter expression to obtain

$$c_{8,s}^{-1}(Lu - f(u))(P) = \int_{\mathbb{R}^8} (u(P) - u(Q))A_P(Q)\,dQ, \qquad A_P(Q) = K(Q-P) - J(Q)K(\Phi(Q) - \Phi(P)). \tag{5.4}$$

Both integrals before subtraction converge absolutely. Near the diagonal this follows from the Lipschitz bound for v in (5.2), the bi-Lipschitz bounds for Φ, and $\sigma < 1$. Away from the diagonal it follows from boundedness.

Two consequences of (5.2) will be used. Write $w_F = 2\chi_F - 1$ and $D_P = |t(\Phi(P))|$. If D_P is large, then $t \circ \Phi$ has the same sign as at P, and has magnitude at least $D_P/2$, throughout $B_{cD_P}(P)$ for a fixed small $c > 0$. This uses the Lipschitz bounds of t and Φ. In this ball, the bound for ∇v and the chain rule control the increments of u by $CD_P^{-\sigma}|h|$ for $|h| \le 1$, and the well bound controls them by $CD_P^{-\sigma}$ for $1 < |h| \le cD_P$. The remaining kernel tail is $O(D_P^{-\sigma})$. For bounded D_P use the global Lipschitz bound and boundedness. Consequently

$$\int_{\mathbb{R}^8} |u(P) - u(P+h)|K(h)\,dh \le C(1 + D_P)^{-\sigma}.$$

Also, for $R \ge 1$,

$$\int_{B_R} |u - w_F|\,dQ \le CR^{8-\sigma}. \tag{5.5}$$

Indeed $\Phi(0) = 0$ and $\Phi(B_R) \subset B_{CR}$ by the uniform Lipschitz bound. Change variables to $\Phi(Q)$, using bounded distortion. The absolute difference is at most $C(1+|t|)^{-\sigma}$ by (5.2). In block polar coordinates the volume density is proportional to $a^3b^3\,d\rho\,dt$, bounded on B_{CR} by $CR^6\,d\rho\,dt$. The ρ-range has length $O(R)$ and $\int_{-CR}^{CR}(1+|t|)^{-\sigma}\,dt = O(R^{1-\sigma})$, giving (5.5).

For the exterior kernel estimates take $r = |P| \geq 3L_1\lambda/2$. Abbreviate $e = e_\lambda(r)$, $\delta = \delta_\lambda(r)$, and $m = m_\lambda(r)$, so $\delta r^{-\sigma} \asymp m$. Take $0 < \theta \leq \theta_0 < 1/6$, with θ_0 to be reduced below. The inequality $r \geq 3L_1\lambda/2$ gives $B_{2\theta r}(P) \subset \{|Q| > L_1\lambda\}$, and $|Q| \asymp r$ on this ball.

Write $h = Q - P$ and $g = \Phi(Q) - \Phi(P)$. The displacement has Lipschitz constant less than $1/4$, so $|g - h| \leq |h|/4$ and the entire segment from h to g stays away from the singularity by at least $3|h|/4$. Also

$$|g - h| \leq C(e_\lambda(r) + e_\lambda(|Q|)), \qquad |J(Q) - 1| \leq C\delta_\lambda(|Q|).$$

The mean value inequality for K then gives, on $|Q - P| > \theta r$,

$$|A_P(Q)| \leq CK(Q-P)\left(\delta_\lambda(|Q|) + \frac{e_\lambda(r) + e_\lambda(|Q|)}{|Q-P|}\right). \tag{5.6}$$

For $0 < b_0 < 1/2$, the part of the integral of $|A_P|$ with $|Q| < b_0 r$ is bounded by $Cb_0^{7-\kappa}m$. In fact $K(Q-P) \leq Cr^{-8-\sigma}$ there, and the radial integrals of $\delta_\lambda(|Q|)$ and $e_\lambda(|Q|)/r$ are bounded by

$$C\lambda^{\kappa+1}(b_0 r)^{7-\kappa}, \qquad C\lambda^{\kappa+1}(b_0 r)^{8-\kappa}/r,$$

respectively, by integrating the bounds $\delta_\lambda(R) \leq \lambda^{\kappa+1}R^{-\kappa-1}$ and $e_\lambda(R) \leq \lambda^{\kappa+1}R^{-\kappa}$ against $CR^7\,dR$. The condition $\kappa < 7$ ensures convergence at zero. The remaining $e_\lambda(r)/r$ term contributes $Cb_0^8 m$. For $B_0 > 2$, the integral with $|Q| > B_0 r$ is at most $CB_0^{-\sigma}m$, since the factor in parentheses in (5.6) is bounded by $C\delta$ there. On $b_0 r \leq |Q| \leq B_0 r$ outside $B_{\theta r}(P)$ one has $|A_P(Q)| \leq C_{\theta,b_0,B_0} m r^{-8}$. In particular $\int_{|Q-P|>\theta r} |A_P(Q)|\,dQ \leq C_\theta m$. We also need an integrated derivative bound for shifts of the center. Differentiating the two kernels in P gives

$$\nabla_P A_P(Q) = -\nabla K(Q-P) + J(Q)D\Phi(P)^{\mathsf{T}}\nabla K(\Phi(Q) - \Phi(P)).$$

The matrix in the second term differs from the identity by

$$\|J(Q)D\Phi(P)^{\mathsf{T}} - I\| \leq C\big(\delta_\lambda(|Q|) + \delta_\lambda(r)\big).$$

Together with $|D^2K| \leq CK(h)/|h|^2$ along the segment from h to g, this gives

$$|\nabla_P A_P(Q)| \leq \frac{CK(h)}{|h|}\left(\delta_\lambda(|Q|) + \delta_\lambda(r) + \frac{e_\lambda(r) + e_\lambda(|Q|)}{|h|}\right).$$

On $|h| > \theta r$ the factor $1/|h|$ is at most $1/(\theta r)$. The term with $\delta_\lambda(r)$ integrates to at most $C_\theta \delta_\lambda(r) r^{-1-\sigma}$, and the other terms are bounded by the same split as above. We obtain

$$\int_{|Q-P|>\theta r} |\nabla_P A_P(Q)|\,dQ \leq C_\theta m/r.$$

Suppose first that $|t(P)| \le \tau\rho(P)$. The functions t and ρ are 1-Lipschitz, and $\rho(P) \ge r/\sqrt{2}$. For $|Q-P| \le 2\theta r$, we have $|t(Q)| \le \tau\rho(P) + 2\theta r$ and $\rho(Q) \ge \rho(P) - 2\theta r$. Thus $\theta_0 < \tau/[2\sqrt{2}(1+2\tau)]$ keeps the ball in $|t| \le 2\tau\rho$, where the transport is exact. Let $\gamma(q) = (q, \eta_\lambda(q))$, let τ_q, N_q be its unit tangent and inward unit normal, and let $k = \eta_\lambda''(1+\eta_\lambda'^2)^{-3/2}$ be its signed curvature. The derivative of N_q in arclength is $-k\tau_q$, by differentiating the unit normal to the graph. Thus the derivatives of $\gamma(q) + dN_q$ with respect to arclength and to d are $(1-dk)\tau_q$ and N_q. The corresponding target derivatives are e_ρ and e_t. Consequently $D\Phi(P)$ sends τ_q to $(1-dk)^{-1}e_\rho$ and N_q to e_t. On the six directions perpendicular to the radial plane, $D\Phi(P)$ multiplies each three-dimensional block by the ratio of the target and source radii. Indeed the radius is constant to first order in an angular direction, and the transport preserves the angular coordinate, so the length of that variation scales by this ratio. Denote these two dilations, extended by the identity on the radial plane, by Λ_P. If O_P is the rotation sending (τ_q, N_q) to (e_ρ, e_t) and fixing the perpendicular directions, then

$$\|D\Phi(P) - O_P\Lambda_P\| \le C\delta|d|/r, \qquad \|\Lambda_P - I\| \le C\delta.$$

Here $|k| \le Ce/r^2$ and the exact region has $|d| \le Cr$. Also $\|O_P - I\| \le C|\eta_\lambda'(q)| \le C\delta$, since the meridional rotation has angle $\arctan(\eta_\lambda'(q))$ in absolute value. Write $h = h_\parallel + h_\perp$ for the two orthogonal projections and put

$$A_{\Lambda_P}(h) = K(h) - \det\Lambda_P\, K(\Lambda_P h).$$

The rotation in the radial plane leaves the kernel and the determinant factor invariant. Taylor expansion of Φ and the Jacobian bound therefore give

$$|A_P(P+h) - A_{\Lambda_P}(h)| \le C\delta\,\frac{|h|+|d|}{r}K(h), \qquad |h| < \theta r. \tag{5.7}$$

Indeed $\Phi(P+h) - \Phi(P) - O_P\Lambda_P h = O(\delta(|h|+|d|)|h|/r)$ by Taylor's formula and the displayed differential estimate. The determinant is locally Lipschitz in matrix entries, so $J(P+h) - \det\Lambda_P = O(\delta(|h|+|d|)/r)$. Both kernel arguments, $\Phi(P+h) - \Phi(P)$ and $O_P\Lambda_P h$, differ from h by at most $C\delta|h|$. The first bound follows from (5.3) on the segment from P to $P+h$, and the second from the bounds on O_P and Λ_P. Thus the segment joining the two arguments stays at length comparable to $|h|$, where $|\nabla K| \le C|h|^{-9-\sigma}$. This proves the estimate, and applying the same inequality to $\Lambda_P - I$ gives $|A_{\Lambda_P}| \le C\delta K$.

For each fixed $h_\parallel \ne 0$, integration of A_{Λ_P} over the perpendicular variables gives zero. This is the linear change of variables $h_\perp \mapsto \Lambda_P h_\perp$ in the second kernel. Let $b_P(h_\parallel) = u(P+h_\parallel)$ on $|h_\parallel| < \theta r$ and $b_P = 0$ off that disk. It follows that

$$\int_{\mathbb{R}^8} \big(u(P) - b_P(h_\parallel)\big) A_{\Lambda_P}(h)\, dh = 0. \tag{5.8}$$

Fubini applies since the difference is $O(|h_\parallel|)$ at zero and bounded elsewhere.

For $0 < R < \theta r$ we claim

$$\frac{1}{|\partial B_R|}\int_{|h|=R} |u(P+h) - b_P(h_\parallel)|\, dS(h) \le CR/r. \tag{5.9}$$

To prove this, fix the first component h_ρ of $h_\parallel$ and write $\ell = h_t$. Each block radius with $h_\perp$ has the form $\sqrt{a_0^2 + |\xi|^2}$, where $a_0 \ge cr$ is the axial radius without $h_\perp$ and $|\xi| \le R$. The identity $\sqrt{a_0^2+|\xi|^2} - a_0 = |\xi|^2/(\sqrt{a_0^2+|\xi|^2} + a_0)$ bounds the radius error by CR^2/r. Since the map on the radii has bounded derivative, the transported radii have the same error. With h_ρ fixed, increasing ℓ in the radial plane decreases the transported first radius and increases the second, at rates bounded away from zero. This follows from $\|D\Phi - I\| < 1/4$ and the untransported rates $-1/\sqrt{2}$ and $1/\sqrt{2}$. Choose $A = C_1R^2/r$ large enough to bracket the radius errors, then shrink θ_0 so that $C_1\theta_0 \le 1$. The planar points with parameters $\ell \pm A$ then stay in $B_{2\theta r}(P)$. Their transported radii bracket those of $\Phi(P+h)$ in the block order. Thus monotonicity bounds $u(P+h)$ between the planar values at $\ell - A$ and $\ell + A$, using actual planar values even when these points leave the disk defining the baseline. The planar function G is monotone with range of length at most two on the slightly larger interval. Its variation measure dG has total mass at most two. Integrating $G(\ell+A)-G(\ell-A)$ in ℓ costs at most $2A\int dG \le 4A$, because each point measured by dG can lie in an interval $[\ell - A, \ell + A]$ for a set of ℓ of length at most $2A$. Given h_ρ, the remaining coordinates are uniform on a sphere of radius $R_1 = (R^2 - h_\rho^2)^{1/2}$ in seven dimensions, so the density of ℓ is bounded by C/R_1. The mean of $1/R_1$ is C/R, since the density of h_ρ/R is proportional to $(1-(h_\rho/R)^2)^{5/2}$. This proves (5.9). The same comparison holds for the transported sign $w_F = 2\chi_F - 1$, since the cone sign has the same coordinate monotonicities.

After subtracting (5.8) from (5.4), the integrand can be written as

$$\begin{aligned}&\big(u(P) - u(P+h)\big)\big(A_P(P+h) - A_{\Lambda_P}(h)\big)\\ &\quad + \big(b_P(h_\parallel) - u(P+h)\big)A_{\Lambda_P}(h).\end{aligned}$$

On $|h| < \theta r$ its integral has absolute value at most

$$Cm\theta^{1-\sigma} + C\frac{\delta|d|}{r}\int_{\mathbb{R}^8} |u(P) - u(P+h)|K(h)\, dh. \tag{5.10}$$

For the $|h|$ term in (5.7), a bounded increment gives $C(\delta/r)\int_0^{\theta r} a^{-\sigma}\, da$. The affine term gives the same integral by (5.9). Both are at most $Cm\theta^{1-\sigma}$. For the last term of (5.10), the increment bound gives $C(\delta|d|/r)(1+|d|)^{-\sigma}$, since the exact map has $D_P = |d|$. This is at most Cm for $|d| \le Cr$, and at most $C_T m r^{\sigma-1} = o(m)$ uniformly when $|d| \le T$. If $|t(P)| > \tau\rho(P)$, then

$$|t(\Phi(P))| \ge \tau r/\sqrt{2} - Ce_\lambda(r) \ge cr$$

by increasing L_0. Here $D_P \ge cr$, so the increment bound gives $O(r^{-\sigma})$. The near kernel is bounded by $C\delta K$, since (5.3) along the segment P to $P+h$ controls the difference of the kernel arguments by $C\delta|h|$ and the

Jacobian error by $C\delta$. This again gives $O(m)$. The far bounds, including $\int_{|h|>\theta r} |A_{\Lambda_P}| \le C_\theta m$, now yield $|Lu - f(u)| \le Cm$ throughout the exterior.

Comparison with the curvature. We record a cutoff estimate used at two scales below. Suppose $G : \mathbb{R}^8 \to \mathbb{R}^8$ is a homeomorphism fixing zero with $\sup_{\overline{B}_R} |G(h) - h| \le \omega < R/2$. Then $B_{R-\omega} \subset G(B_R) \subset B_{R+\omega}$. The outer inclusion is immediate. For the inner inclusion, a segment from zero to a point in $B_{R-\omega}$ cannot meet $G(\partial B_R)$, whose points have norm at least $R-\omega$. Thus the symmetric difference of the image and B_R lies in a shell of volume $O(R^7\omega)$. After applying an invertible linear map with uniformly bounded norm and inverse norm, the shell remains at distances comparable to R and its volume is still $O(R^7\omega)$. Its integral against K is therefore $O(\omega R^{-1-\sigma})$.

Let $P = p + d\nu(p)$, $|d| \le T$, and $r = |P| \ge 2L_1\lambda$. Then $|p| \ge r - T \ge 3L_1\lambda/2$ for all sufficiently large λ. The points p, P have the same block directions. At p, $|t(p)| = \eta_\lambda(\rho(p)) \le \tau\rho(p)/4$ by the choice of L_0. Since t, ρ change by at most T from p to P, both points lie in the smaller region $|t| \le \tau\rho$ for large λ. Since $|p| \asymp r$, the near estimates apply at p with the same truncation radius θr after choosing θ_0 smaller if needed. At p the corresponding sharp identity is

$$H_\sigma[F](p) = \mathrm{PV} \int_{\mathbb{R}^8} -w_F(Q) A_p(Q)\, dQ. \tag{5.11}$$

To justify (5.11), truncate both kernels by excluding $B_\varepsilon(p)$ in the Q variables. In the transformed term, change variables to $\Phi(Q)$. The sign becomes that of D, and the deleted set is $\Phi(B_\varepsilon(p))$. Center at $\Phi(p)$ and apply the cutoff estimate with

$$G(h) = D\Phi(p)^{-1}(\Phi(p+h) - \Phi(p)), \qquad \omega = O(\varepsilon^2)$$

by Taylor's formula. After applying $D\Phi(p)$, this replaces the deleted set by the centered differential ellipsoid at cost $O(\varepsilon^{1-\sigma})$. The difference between truncation by this ellipsoid and by a centered ball is supported in a ball of radius $C\varepsilon$. The tangent-plane sign integrates to zero for both truncations by central reflection. For the difference between the cone and tangent signs, (3.5) bounds the cost by $O(\varepsilon^{1-\sigma})$, since $\Phi(p) \in \Sigma \setminus \{0\}$. The cone integral with a centered spherical truncation is zero by Section 3. Letting $\varepsilon \downarrow 0$ gives (5.11).

Use the sharp planar baseline $b_p^\sharp(h_\parallel) = w_F(p + h_\parallel)$ for $|h_\parallel| < \theta r$, and zero outside. Its principal value integral against A_{Λ_p} vanishes. To see this, subtract the sign of the tangent line in the radial plane. This sign is odd and its integral against the even affine kernel vanishes on every spherical truncation. The remaining difference depends only on $h_\parallel$. For small $h_\parallel$, the sharp baseline is the cap sign in the plane of the two radii, because F_λ equals E_λ near p. Write $h_\parallel = (h_\rho, h_t)$ in that plane. Taylor expansion of the graph defining function at the radial coordinates of p gives

$$h_t - \big(\eta_\lambda(\rho(p) + h_\rho) - \eta_\lambda(\rho(p))\big) = \sqrt{1 + \eta_\lambda'(\rho(p))^2}\, \nu(p) \cdot h_\parallel + O(|h_\parallel|^2).$$

Thus the signs can differ only where $|\nu(p)\cdot h_\parallel| \le C|h_\parallel|^2$. Since $|h_\parallel| \le |h|$, the local interface estimate in (3.5) applies to this slab and gives absolute integrability against $K(h)$. At infinity use boundedness and the kernel tail. Fubini and the zero perpendicular marginal of A_{Λ_p} give zero for the remaining difference. Thus the sharp near integrand after subtraction can be written as

$$-w_F(p+h)\big(A_p(p+h)-A_{\Lambda_p}(h)\big)+\big(b_p^\sharp(h_\parallel)-w_F(p+h)\big)A_{\Lambda_p}(h).$$

The first term has integral bounded by $Cm\theta^{1-\sigma}$ by (5.7) at $d=0$. The second has the same bound by (5.9) for w_F.

The diffuse far integral has three terms. Its integrand is

$$\begin{aligned}&u(P)\big(A_P(P+h)-A_{\Lambda_P}(h)\big)\\&\quad-u(P+h)A_P(P+h)+b_P(h_\parallel)A_{\Lambda_P}(h).\end{aligned}$$

The sharp far integrand is

$$-w_F(p+h)A_p(p+h)+b_p^\sharp(h_\parallel)A_{\Lambda_p}(h).$$

To bound the coefficient of $u(P)$ in the diffuse far integral, observe that the $K(h)$ terms in $A_P(P+h)-A_{\Lambda_P}(h)$ cancel. Changing variables in the two remaining kernel terms writes the coefficient as the difference of integrals of K over the complements of the centered image of $B_{\theta r}(P)$ and the affine ellipsoid $\Lambda_P B_{\theta r}$. Apply the cutoff estimate with $R=\theta r$ and $G(h)=\Lambda_P^{-1}O_P^{-1}(\Phi(P+h)-\Phi(P))$. The Taylor and radial-plane stretch estimates give $\omega = C\delta(R^2+|d|R)/r$. For small θ_0 and large λ on the fixed normal band, $\omega < R/2$. The linear map $O_P\Lambda_P$ sends B_R to the affine ellipsoid, since the radial rotation leaves that ellipsoid invariant. The shell bound $C\omega R^{-1-\sigma}$ therefore gives the following bound for the coefficient

$$Cm\left(\theta^{1-\sigma}+\frac{|d|}{r}\theta^{-\sigma}\right). \tag{5.12}$$

For the far term with A_P, compare the two centers on the common integration region in the Q variables. For sufficiently large λ, each intermediate center on the segment from P to p has norm r' between $r-T$ and $r+T$. Its distance from Q is at least $\theta r/2$, hence at least $\theta r'/4$. Apply the integrated derivative estimate at this center with cutoff $\theta r'/4$. Since $m_\lambda(r')/r' \asymp m/r$, integrating along the segment gives an error $C_\theta m|d|/r$. The two truncations differ only in a shell about P of radius θr and thickness at most $2|d|$, of volume $O((\theta r)^7|d|)$. On this shell $|Q| \asymp r$ and both center distances are comparable to θr. The pointwise bound $C_\theta m r^{-8}$ gives the same cost there.

To replace u by w_F, the portions $|Q|<b_0 r$ and $|Q|>B_0 r$ cost at most $Cm(b_0^{7-\kappa}+B_0^{-\sigma})$. On the remaining annulus, (5.5) with $R=B_0 r$ bounds the L^1 error by $C_{B_0}r^{8-\sigma}$. Multiplication by the pointwise far kernel bound gives a cost $C_{\theta,b_0,B_0}mr^{-\sigma}$.

For the baseline term use the same radial plane and the same disk cutoff at p and P. On the common perpendicular subspace the dilations are the

restrictions of $D\Phi$, so the second derivative bound in (5.3) along the segment gives $\|\Lambda_P - \Lambda_p\| \le C\delta|d|/r$. The determinant and kernel difference then cost $C\delta(|d|/r)K(h)$ pointwise by the mean value inequality. Integration on $|h| > \theta r$ gives $C_\theta m|d|/r$. Put $t_P(h_\parallel) = t(\Phi(P + h_\parallel))$ on the disk. Both block radii stay positive and the block directions stay fixed on this planar slice. Thus t has gradient e_t there, the unit vector for the second meridional coordinate. The chain rule and $\|D\Phi - I\| < 1/4$ give

$$\partial_{h_t} t_P = e_t \cdot D\Phi(P + h_\parallel)e_t \ge 3/4.$$

Moreover $|t_P - t_p| \le C|d|$ by the Lipschitz bound on $t \circ \Phi$. Consequently

$$|b_P - b_p^\sharp| \le C(1 + |t_P|)^{-\sigma} + 2\mathbf{1}_{\{|t_P| \le C|d|\}}$$

on the disk. For each fixed h_ρ, change variables from h_t to t_P. Since $\Phi(0) = 0$ and Φ is uniformly Lipschitz, the range of t_P lies in $[-Cr, Cr]$. The two terms then integrate to at most $Cr^{1-\sigma}$ and $C|d|$, respectively. The h_ρ interval has length $O(r)$, giving $\int |b_P - b_p^\sharp|\, dh_\parallel \le C(r^{2-\sigma} + |d|r)$. For $|h_\parallel| < \theta r$ one has

$$\int_{\{|h| > \theta r\}} K(h_\parallel + h_\perp)\, dh_\perp \le C(\theta r)^{-2-\sigma}.$$

If $|h_\parallel| \ge \theta r/2$, integrate over all six perpendicular variables and scale by $|h_\parallel|$. If $|h_\parallel| < \theta r/2$, the integration requires $|h_\perp| \ge \sqrt{3}\theta r/2$, and the bound follows by radial integration in six dimensions. Since $|A_{\Lambda_p}| \le C\delta K$, the baseline replacement costs at most $C_\theta m(r^{-\sigma} + |d|/r)$.

Together these estimates give

$$\frac{|c_{8,s}^{-1}(Lu_\lambda - f(u_\lambda))(P) - H_\sigma[F_\lambda](p)|}{m} \le C\big(\theta^{1-\sigma} + b_0^{7-\kappa} + B_0^{-\sigma}\big) + C_{\theta,b_0,B_0,T}\big(r^{-\sigma} + r^{-1} + r^{\sigma-1}\big).$$

The first constant is independent of θ, b_0, B_0. For fixed θ, b_0, B_0, T, the second term tends to zero uniformly on the band since $r \ge 2L_1\lambda$ and $0 < \sigma < 1$. Let $b_0 \downarrow 0$, $B_0 \uparrow \infty$, and then $\theta \downarrow 0$. This gives

$$(Lu_\lambda - f(u_\lambda))(P) = c_{8,s} H_\sigma[F_\lambda](p) + o(m_\lambda(r))$$

uniformly on each stated band.

Changing the interior data. Replacing u_λ by V_λ changes the residual at P by

$$c_{8,s} \int_{B_{L_1\lambda}} (u_\lambda - V_\lambda)(Q) K(Q - P)\, dQ.$$

The reaction is unchanged at P, and the kernel integral is well defined because $V_\lambda = u_\lambda$ near P and the core data are bounded and measurable. For Q in the core, $|Q - P| \ge r/2$. Thus the residual change is bounded by $C_{L_1}\lambda^8 r^{-8-\sigma}$, which is $O(m_\lambda(r))$ since $\lambda^8 r^{-8-\sigma}/m_\lambda(r) = (\lambda/r)^{7-\kappa}$. This proves the second assertion.

On the fixed normal band, p is outside the core for large λ, so F_λ and E_λ agree near p. Write $w_A = 2\chi_A - 1$ for either set. The difference of their curvatures is an ordinary integral on the core. More precisely, with $\mathcal{R}(w) = Lw - f(w)$, we have

$$\begin{aligned}&\mathcal{R}(V_\lambda)(P) - c_{8,s}H_\sigma[E_\lambda](p)\\&\quad - \big(\mathcal{R}(u_\lambda)(P) - c_{8,s}H_\sigma[F_\lambda](p)\big)\\&= c_{8,s}\int_{B_{L_1\lambda}} \Big[(u_\lambda - V_\lambda)(Q)K(Q-P) - (w_{F_\lambda} - w_{E_\lambda})(Q)K(Q-p)\Big]\,dQ.\end{aligned}$$

For Q in the core, its distance to any point on the segment from P to p is at least $r/2 - T \geq r/3$ for large λ. The mean value bound for K shows that replacing $K(Q-p)$ by $K(Q-P)$ costs at most $C_{L_1,T}\lambda^8 r^{-9-\sigma}$. Dividing by $m_\lambda(r)$ bounds this by $C_{L_1,T}(\lambda/r)^{7-\kappa}/r \to 0$ uniformly on the band.

With the same kernel, the difference of the two terms is $(u_\lambda - w_{F_\lambda}) - (V_\lambda - w_{E_\lambda})$. Its L^1 norm on the core is $o(\lambda^8)$. The first difference is $O_{L_1}(\lambda^{8-\sigma})$ by (5.5), and the second is $o(\lambda^8)$ by the assumed core convergence. Multiplying by the bound $Cr^{-8-\sigma}$ for the core kernel gives $o(m_\lambda(r))$, by the same ratio as above. This proves (5.1).

Finally, the curvature assumption and (3.4) give

$$H_\sigma[E_\lambda](p) \leq -c\lambda^{-\sigma}(1 + |p|/\lambda)^{-\kappa-1-\sigma} \leq -c' m_\lambda(r).$$

Indeed $r \geq 2L_1\lambda$ and $|p| \leq r + T \leq 2r$ for large λ, so $c' > 0$ can be chosen independently of T. Absorbing the uniform error for sufficiently large λ gives the strict residual bound. □

6. The diffuse minimizer in eight dimensions

Proposition 7. *The cap (3.1), with the geometric input (3.2)–(3.3), yields a nonconstant entire minimizer $v : \mathbb{R}^8 \to (-1, 1)$ of (2.1). The minimizer is smooth and invariant under reflection of every coordinate.*

Write the saddle from Section 4 as $S(\rho, t)$. We use its block monotonicity, its strict increase in t on $(-\rho, \rho)$, and its radial monotonicity $S_\rho \geq 0$ for $t > 0$. Lemma 5 supplies the estimate

$$|S_\rho(\rho, t)| \leq C/\rho \qquad (|t| < \rho). \tag{6.1}$$

6.1. **Gluing the profiles.** Let d_λ denote the signed distance to ∂E_λ, positive in E_λ, and set $\mathcal{R}(V) = LV - f(V)$. The signed distance is 1-Lipschitz. For two points on the same side this follows from the ordinary distance bound, and for points on opposite sides the segment between them meets the boundary, so the sum of their distances is at most the length of the segment.

Apply Proposition 6 with $v = S$ and $E_1 = E$. The hypotheses follow from Sections 3 and 4. Take L_0 to be an exterior radius for its map Φ_λ. On the cap with $|p| \geq L_0\lambda$ the arclength label is comparable to $\rho(p)$ by Proposition 6, since $\rho(p) \geq |p|/\sqrt{2}$. The height derivative $\eta'_\lambda(\rho(p))$ is negative off the axis

by the formula for η' in Section 3. Then fix $L_1 > 4L_0$. The map is the same for all $L_1 \geq L_0$, so its exact tube and the equality $\Phi_\lambda^{-1}(D) = E_\lambda$ apply beyond $L_0\lambda$, including in the blending annulus below.

On $\partial E \cap \overline{B}_{4L_1}$, compactness and the tubular-neighborhood theorem for a slightly larger portion of the smooth cap give a common tubular radius $4\delta > 0$. The boundary outside the larger portion cannot be a closest point within this tube if δ is small. After dilation, points in the tube have a unique smooth footpoint $p = p(P)$ and $P = p + d_\lambda(P)\nu(p)$, with ν the inward normal. The scaled distance and footpoint are obtained by applying the unit-scale maps at P/λ and multiplying by λ. Hence $D^2 d_\lambda = O(\lambda^{-1})$ and the first derivative of $p(P)$ is uniformly bounded. By the separation in Section 3, we can take the tube inside D. We also require $4\delta < \operatorname{dist}(0, \overline{E})$, which is positive by the geometry of the cap.

On this compact portion of the dilated cap, define a smooth radial label $r_\lambda(p)$. Take it equal to $c_1\lambda$ for $|p| \leq L_0\lambda$ and to the arclength label $S_\lambda(\rho(p))$ of Proposition 6 for $|p| \geq 2L_0\lambda$, using a smooth convex interpolation in $|p|/\lambda$. The label satisfies $r_\lambda(p) = \lambda r_1(p/\lambda)$. On the compact scaled portion it is therefore comparable to λ, with uniformly bounded first derivative.

Where the arclength label has positive weight, $|p| \geq L_0\lambda$. By taking L_0 large we have $\eta_\lambda(\rho(p)) \leq \rho(p)/2$. Both block radii at p are therefore at least $\rho(p)/(2\sqrt{2}) \geq L_0\lambda/4$. Reduce δ so that $4\delta < L_0/8$. Each radius is 1-Lipschitz, so neither can vanish along the normal segment of length at most $4\delta\lambda$. The block directions stay fixed on this segment. The inward normal's radial component is $-\eta_\lambda'/\sqrt{1 + (\eta_\lambda')^2} \geq 0$, so $\rho(P) \geq \rho(p)$ for a positive normal displacement. Since $S_\lambda(\rho(p)) \leq \rho(p)$, the arclength label is at most $\rho(P)$. The constant label has the same property for sufficiently small $c_1 > 0$. In fact $\rho(p) \geq \lambda/\sqrt{2}$ on the cap and ρ changes by at most $4\delta\lambda$ in the tube. We can take $4\delta < 1/(2\sqrt{2})$ and $c_1 < 1/(2\sqrt{2})$. Thus $r_\lambda(p) \leq \rho(P)$ when $d_\lambda(P) \geq 0$. After fixing the labels, reduce δ so that $4\delta\lambda < r_\lambda(p)$ throughout the compact tube and $\delta < L_1/2 - 2L_0$. Also require $\delta < 2cL_0$ for the tube constant c in Proposition 6.

For $|P| \leq 3L_1\lambda$, define an inner profile I_λ as follows. Use $S(r_\lambda(p), d_\lambda(P))$ when $|d_\lambda| \leq \delta\lambda$. In $\delta\lambda < |d_\lambda| < 2\delta\lambda$, blend it with the phase background using a fixed smooth cutoff of d_λ/λ. Use the background alone beyond this transition. The background is -1 for $d_\lambda < 0$ and $1 - C_1\lambda^{-\sigma}$ for $d_\lambda > 0$. The weight of the normal profile is one near $d_\lambda = 0$ on both sides. Thus the background jump at the cap is absent from I_λ. Where the normal profile is used, the signed distance and footpoint are smooth. Outside this tube the formula is locally constant on each phase. The smooth cutoff joins these formulas within the tube. When a footpoint is needed, its radius is at most $3L_1\lambda + 2\delta\lambda < 4L_1\lambda$, and the profile argument satisfies $|d_\lambda| < r_\lambda(p)$.

In the positive phase of this ball, the separation from the cone in Section 3 gives $t(P) \geq c\lambda$. The well estimate then gives $1 - S(P) \leq C\lambda^{-\sigma}$. Choose

$C_1 > C$, and take λ sufficiently large that the positive background belongs to $(-1, 1)$. Both backgrounds are strictly below $S(P)$ where used.

The normal profile is strictly below $S(P)$. For nonpositive normals this follows from $P \in D$ and the nonpositive profile value. On a positive normal, $P \in E_\lambda$. A nearest cone point is outside $\overline{E}_\lambda$ by the strictly positive right-hand side in (3.1). The segment to this point crosses ∂E_λ before the endpoint, so $d = d_\lambda(P) < t(P)$. With the bound $r_\lambda(p) \le \rho(P)$ and $S_\rho \ge 0$, we have

$$S(r_\lambda(p), d) \le S(\rho(P), d) < S(\rho(P), t(P)).$$

For the strict inequality, choose an intermediate normal coordinate between d and $t(P)$ inside $(-\rho(P), \rho(P))$ and use strict increase up to that coordinate. This also covers the case $t(P) = \rho(P)$ on an axis.

Take a smooth radial cutoff $\xi_\lambda(P)$ equal to one for $|P| \le L_1\lambda/2$ and zero for $|P| \ge L_1\lambda$, with values in $[0, 1]$ and depending on $|P|/\lambda$. Define V_λ as the blend of I_λ with $S \circ \Phi_\lambda$, using ξ_λ for the inner weight. Beyond its support use the transported saddle alone. Then V_λ is locally Lipschitz and $-1 \le V_\lambda \le S$, by the inner comparison and Proposition 6. It is jointly continuous in (λ, P), since the geometry maps and cutoffs are scaled from fixed maps and the background depends continuously on λ.

For $|P| \ge L_1\lambda/2$ in the compact tube and $|d_\lambda| \le \delta\lambda$, the footpoint satisfies $|p| \ge L_1\lambda/2 - \delta\lambda > 2L_0\lambda$. The choice of δ ensures $|d_\lambda| \le c|p|$. Thus the exact transport uses the same label and normal distance as the inner profile. In particular,

$$V_\lambda(P) = S(r_\lambda(p), d_\lambda(P)) \qquad (|P| \le 3L_1\lambda,\ |d_\lambda(P)| \le \delta\lambda). \tag{6.2}$$

This identity also covers the pure transport part of the compact tube beyond the blending annulus. Where the exterior transport is exact, it strictly decreases the saddle value. Its first block radius increases and its second decreases. Along the segment between the two radial pairs the radii stay positive and the block derivative signs from Section 4 give strict decrease off the cone, which this segment can meet at most once.

In the blend and the exterior,

$$|t(\Phi_\lambda(P))| \ge c\,|d_\lambda(P)| \qquad (|P| \ge L_1\lambda/2). \tag{6.3}$$

Indeed Φ_λ takes $\partial(\Phi_\lambda^{-1}D)$ to Σ, so

$$|t(\Phi_\lambda(P))| = \inf_{Q \in \partial(\Phi_\lambda^{-1}D)} |\Phi_\lambda(P) - \Phi_\lambda(Q)| \ge c \inf_{Q \in \partial(\Phi_\lambda^{-1}D)} |P - Q|$$

by the Lipschitz bound for the inverse. A nearest point on the preimage boundary in $\{|Q| > L_0\lambda\}$ lies on ∂E_λ, since the two sets agree on that open exterior region. If the nearest point lies in $\{|Q| \le L_0\lambda\}$, its distance to P is at least $|P| - L_0\lambda \ge |P|/2$. This also bounds the cap distance up to a constant, since for a fixed $p_* \in \partial E$, $|d_\lambda(P)| \le |P - \lambda p_*| \le C|P|$. The signs of $t(\Phi_\lambda(P))$ and $d_\lambda(P)$ agree in this region.

For $\varepsilon \in \{-1, 1\}$ and $\varepsilon d_\lambda(P) > 0$, the well estimate, the background values and (6.3) give the single bound

$$|V_\lambda(P) - \varepsilon| \le C(1 + |d_\lambda(P)|)^{-\sigma} + C\lambda^{-\sigma}. \tag{6.4}$$

For the inner profile, this is the saddle estimate at signed cone distance $d_\lambda(P)$, and on the blend it follows by convexity of the bound. In particular $V_\lambda(\lambda X) \to 2\mathbf{1}_E(X) - 1$ off ∂E. This holds also for a sequence of scaled points converging off ∂E, because $d_\lambda(\lambda X) = \lambda d_1(X)$. The boundary has measure zero, so bounded convergence after scaling gives the core L^1 hypothesis of Proposition 6.

6.2. **Residual on bounded scaled regions.** Suppose $|P| \le 2L_1\lambda$ and $|d_\lambda(P)| \le \delta\lambda/4$, and write $P = p + d\nu(p)$. Put $r_0 = r_\lambda(p)$ and define $F(z) = S(r_0, z)$, extending it by the endpoint values for $|z| \ge r_0$. Choose a cone normal comparison point X_0 with radial coordinates (r_0, d) and both block directions equal to a fixed unit vector in $\mathbb{R}^4$. Write ν_0 for the inward cone normal in these directions. By (6.2), $V_\lambda(P) = S(X_0) = F(d)$. In the residual, we can therefore replace $f(V_\lambda(P))$ by $LS(X_0)$. The two operator integrals have the same center value $F(d)$. The planar profiles $F(d+\nu(p)\cdot h)$ and $F(d + \nu_0 \cdot h)$ give equal operator values at $h = 0$, since the kernel is rotation invariant. Subtracting these equal planar values from the two operator integrals yields

$$\begin{aligned} c_{8,s}^{-1}\mathcal{R}(V_\lambda)(P) &= \int_{\mathbb{R}^8} [D_\lambda(h) - D_\lambda^0(h)]|h|^{-8-\sigma}\, dh, \\ D_\lambda(h) &= F(d + \nu(p)\cdot h) - V_\lambda(P + h), \\ D_\lambda^0(h) &= F(d + \nu_0 \cdot h) - S(X_0 + h). \end{aligned} \tag{6.5}$$

The profiles are locally Lipschitz at the evaluation points and the integrals converge absolutely since $\sigma < 1$.

Choose $b_0 > 0$ small enough that $b_0 \le \min(\delta/4, L_1/2)$. Then, for $|h| = a \le b_0\lambda$, the segment from P to $P + h$ remains in the region of (6.2), by the distance and radius bounds. The derivative of the footpoint and label maps gives a label change of at most Ca. The same estimate holds for the cone comparison since ρ is 1-Lipschitz. For b_0 sufficiently small, the cone comparison stays away from both axes. In both comparisons, the initial label satisfies $r_0 > 4\delta\lambda$, the label changes by at most Ca, and the absolute normal coordinate is at most $|d| + a \le \delta\lambda/2$. By reducing b_0 so that $Cb_0 < \delta$, every intermediate label remains larger than the absolute normal coordinate and comparable to λ. Holding the normal coordinate fixed and replacing the variable label by r_0 thus costs $O(a/\lambda)$, by (6.1). In the cap tube $\nabla d_\lambda(P) = \nu(p)$ and $\|D^2 d_\lambda\| \le C/\lambda$. The cone distance has gradient ν_0 at X_0 and the same Hessian bound nearby: the two radii of X_0 are $(r_0 - d)/\sqrt{2}, (r_0 + d)/\sqrt{2}$, each comparable to λ, and each block norm has Hessian bounded by the reciprocal of its radius. Thus the distance differs from its planar argument by $O(a^2/\lambda)$.

To bound the effect of this last error without losing a power of λ, use the variation measure μ of the increasing function F. Its mass is at most two. The profile error from the distance is bounded by

$$\mu\big([d+au-Ca^2/\lambda,\ d+au+Ca^2/\lambda]\big), \qquad u=\nu\cdot h/a,$$

where ν is the corresponding unit normal. On the unit sphere in $\mathbb{R}^8$, the density of u is proportional to $(1-u^2)^{5/2}$ on $[-1,1]$ and is bounded. For any fixed point in the support of μ, the interval of u for which it belongs to the displayed interval has length at most $2Ca/\lambda$. Fubini against the normalized angular measure therefore gives an angular average at most Ca/λ for this error. Including the radial label error, the contribution from $|h|<b\lambda$ in (6.5) for $0<b\le b_0$ is at most

$$C\int_0^{b\lambda}(a/\lambda)a^{-1-\sigma}\,da = C'b^{1-\sigma}\lambda^{-\sigma}.$$

The outer integral is bounded by $C(b\lambda)^{-\sigma}$ by boundedness. Taking $b=b_0$ gives the absolute residual estimate.

Now fix M and restrict to $|d|\le M$. Along any sequence $\lambda\to\infty$ and such evaluation points, extract a subsequence so that $p/\lambda\to p_\infty$, $\nu(p)\to\nu_\infty$, and $r_0/\lambda\to r_\infty>0$. Then $P/\lambda\to p_\infty$ and $X_0/\lambda\to X_\infty\in\Sigma\setminus\{0\}$. After putting $h=\lambda k$, the values $V_\lambda(P+\lambda k)$ converge to the cap sign at $p_\infty+k$ away from ∂E by (6.4). Similarly $S(X_0+\lambda k)$ converges to the cone sign at $X_\infty+k$ off Σ. When the limiting planar normal component of k is nonzero, the argument of F has that sign and magnitude at least $c(k)\lambda$. Clamping the argument to $[-r_0,r_0]$ preserves the sign and a lower bound of the same order, since $r_0/\lambda\to r_\infty>0$. The clamped argument is exactly the cone distance in the saddle profile defining F. The well estimate of Lemma 4 therefore gives convergence of the planar profiles to their half-space signs. The exceptional interfaces and planes have measure zero. On $|k|\ge b$ dominated convergence applies with majorant $C|k|^{-8-\sigma}$. The planar signs integrate to zero on this region by reflection. The centered cone truncation at X_∞ is zero by Section 3. Thus the only remaining term in the limit of the truncated integral is the negative cap sign, which gives the exterior-minus-interior curvature truncated at b.

For fixed $b>0$, the truncated cap curvature depends continuously on the footpoint by dominated convergence in variables centered there, since ∂E has measure zero. Estimate (3.5) gives an error $O(b^{1-\sigma})$ for the deleted curvature integral. Its constant is uniform on the compact set of footpoints by the Hessian bound for the unit signed distance. Thus the curvature is continuous there as a uniform limit of the truncated integrals. The deleted diffuse terms have the same bound after multiplication by λ^σ. Letting $b\downarrow 0$ yields the uniform limit

$$\lambda^\sigma\mathcal{R}(V_\lambda)(P)=c_{8,s}H_\sigma[E](p/\lambda)+o(1) \qquad (|d|\le M,\ |P|\le 2L_1\lambda). \tag{6.6}$$

Indeed, any sequence violating uniform convergence would admit a subsequence of the type just considered. Continuity of the curvature would then contradict the violation.

If $|d_\lambda(P)| \geq \delta\lambda/4$, a ball of radius $c\lambda$ about P, with $c \leq \min(\delta/8, L_1/2)$, lies in the same phase and has cap distance at least $\delta\lambda/8$, by the Lipschitz bound for signed distance. It also stays within $|Q| < 3L_1\lambda$, where the compact profile is defined. The well error is $O(\lambda^{-\sigma})$ there by (6.4). We claim the Lipschitz bound on that ball is also $O(\lambda^{-\sigma})$. For the inner profile, its ρ, t derivatives are directional derivatives of S in an orthonormal radial plane. They have this bound by Lemma 4, since the distance argument is at least $\delta\lambda/8$ in absolute value. The label and normal coordinate have bounded derivatives where this profile is used. For $S \circ \Phi_\lambda$, use (6.3), the same well estimate, and the uniform bound on $D\Phi_\lambda$. The backgrounds are constant in the ball's phase. In each convex combination, derivatives of the cutoffs multiply differences between two values within $O(\lambda^{-\sigma})$ of the same well, so they preserve the derivative bound. Integrating along each segment in the ball gives the claimed Lipschitz bound there. For $LV_\lambda(P)$, integrate the Lipschitz increment at distances below one, twice the well error between one and $c\lambda$, and bounded increments in the tail, which costs $O(\lambda^{-\sigma})$. The reaction is $O(\lambda^{-\sigma})$ as f vanishes at that well. This gives the same absolute residual bound throughout the rest of the compact scaled region.

6.3. **Correction in the pure phases.** Put $\beta = \kappa + 1 + \sigma < 8$ and

$$\widetilde{m}_\lambda(P) = \lambda^{\beta-\sigma}(\lambda^2 + |P|^2)^{-\beta/2}.$$

On $|P| \leq 2L_1\lambda$ this weight is comparable to $\lambda^{-\sigma}$. On $|P| \geq 2L_1\lambda$ it is comparable to the exterior weight $\lambda^{\kappa+1}|P|^{-\kappa-1-\sigma}$ of Proposition 6, since $\beta - \sigma = \kappa + 1$ and $(\lambda^2 + |P|^2)^{1/2} \asymp |P|$ there.

The compact absolute residual bound and the second assertion of Proposition 6 give the global absolute bound below. For the sign on a fixed band $|d_\lambda| \leq M$, first consider $|P| \leq 2L_1\lambda$. Its footpoint is in $|p| \leq 3L_1\lambda$ for large λ, and (3.3) bounds the curvature at p/λ above by a strictly negative constant on this compact set. Thus (6.6) gives a negative bound of order $\lambda^{-\sigma}$ with a coefficient independent of M. For $|P| \geq 2L_1\lambda$, choose a nearest cap point p, which exists by closedness. Minimizing squared distance along the smooth boundary makes $P - p$ normal at p. If $P \neq p$, the open segment from p to P has no boundary point, so the normal displacement has the same sign as $d_\lambda(P)$. Since $|p| \geq |P| - M$, this displacement lies in the exact tube of Proposition 6 for large λ. Its signed residual bound applies by (3.3) and the equivalent bound in $|p|$ from Section 3. Combining the two regions gives

(6.7)
$$|\mathcal{R}(V_\lambda)| \leq C_{\mathrm{res}}\widetilde{m}_\lambda \quad \text{everywhere}, \qquad \mathcal{R}(V_\lambda) \leq -c_{\mathrm{res}}\widetilde{m}_\lambda \quad \text{on } |d_\lambda| \leq M.$$

The constants are independent of M, and the lower threshold on λ for the signed estimate may depend on M. Also (6.4) implies $|V_\lambda| \geq a_0 > 1/\sqrt{3}$ when $|d_\lambda| \geq T_0$, for a fixed sufficiently large T_0 and sufficiently large λ.

Take a smooth even $0 \leq \chi \leq 1$ with $\chi = 0$ on $[-1,1]$ and $\chi = 1$ outside $(-2,2)$. For $\lambda \geq T$ one has

$$\left|L(\widetilde{m}_\lambda \chi(d_\lambda/T))(P)\right| \leq CT^{-\sigma}\widetilde{m}_\lambda(P). \tag{6.8}$$

Write $R_P = (\lambda^2+|P|^2)^{1/2}$. The signed-distance Lipschitz bound controls the cutoff increment by $C\min(1,|P-Q|/T)$. For $|P-Q| \leq R_P/4$, the quantity R_Z is comparable to R_P on the segment from P to Q, and $|\nabla \widetilde{m}_\lambda(Z)| \leq C\widetilde{m}_\lambda(P)/R_P$ there. The product increment is therefore bounded by

$$C\widetilde{m}_\lambda(P)\left(\frac{|P-Q|}{R_P} + \min(1,|P-Q|/T)\right).$$

Integrating the first term against the radial kernel gives $CR_P^{-\sigma}\widetilde{m}_\lambda(P)$. For the second, the radial integral

$$\int_0^\infty \min(1,r/T)\, r^{-1-\sigma}\, dr = \left(\frac{1}{1-\sigma} + \frac{1}{\sigma}\right) T^{-\sigma}$$

gives the required cutoff bound.

At distances $|P-Q| \geq R_P/4$, bound the increment by $\widetilde{m}_\lambda(P) + \widetilde{m}_\lambda(Q)$. The first weight contributes $CR_P^{-\sigma}\widetilde{m}_\lambda(P)$ by the kernel tail. For the second weight on $|Q| \leq 2R_P$, bound the kernel by $CR_P^{-8-\sigma}$ and use

$$\int_{|Q|\leq 2R_P} \widetilde{m}_\lambda(Q)\, dQ \leq C\lambda^{\beta-\sigma} R_P^{8-\beta},$$

which follows by integrating $\lambda^{\beta-\sigma}|Q|^{-\beta}$ and using $\beta < 8$. On $|Q| > 2R_P$, use $|P-Q| \geq |Q|/2$ to bound the integral by $C\lambda^{\beta-\sigma}\int_{2R_P}^\infty r^{-1-\beta-\sigma}\, dr$. Both estimates give $CR_P^{-\sigma}\widetilde{m}_\lambda(P)$. Since $R_P \geq \lambda \geq T$, the near and far bounds prove (6.8).

Choose $1/\sqrt{3} < a_1 < a_0$ and $c_w = 3a_1^2 - 1 > 0$. First choose A so $c_w A > 4C_{\mathrm{res}}$. Then choose $T \geq T_0$ so that $C_{\mathrm{cut}} AT^{-\sigma} < \min(c_{\mathrm{res}}, c_w A)/2$, with C_{cut} from (6.8). These choices are possible because the residual constants are independent of the band width. Finally take λ large enough for the signed estimate on $|d_\lambda| \leq 2T$ and for $\lambda \geq T$. For large λ, the shift $\zeta = A\widetilde{m}_\lambda \chi(d_\lambda/T)$ is at most $a_0 - a_1$, since $\widetilde{m}_\lambda \leq \lambda^{-\sigma}$. If $\zeta(P) > 0$ then $|d_\lambda(P)| \geq T_0$. The whole segment from $V_\lambda(P) - \zeta(P)$ to $V_\lambda(P)$ lies where $|a| \geq a_1$, so $f'(a) \leq -c_w$ on that segment. The mean value bound $f(V_\lambda) - f(V_\lambda - \zeta) \leq -c_w\zeta$ follows and also holds where $\zeta = 0$. Consequently

$$\mathcal{R}(V_\lambda - \zeta) \leq \mathcal{R}(V_\lambda) + C_{\mathrm{cut}} AT^{-\sigma}\widetilde{m}_\lambda - c_w\zeta.$$

For $|d_\lambda| \leq 2T$, use (6.7) with $M = 2T$ to bound this by $-c_{\mathrm{res}}\widetilde{m}_\lambda/2$. For $|d_\lambda| > 2T$, we have $\zeta = A\widetilde{m}_\lambda$ and the bound is at most $(C_{\mathrm{res}} - c_w A/2)\widetilde{m}_\lambda < 0$. This proves the strict subsolution inequality before clipping. Set

$$a_\lambda = \max\{-1, V_\lambda - \zeta\}. \tag{6.9}$$

Where $a_\lambda > -1$, clipping raises other values and can only decrease the fractional Laplacian. Where $a_\lambda = -1$, its Laplacian is nonpositive by the minimum property. Thus a_λ is a pointwise subsolution, strict off its plateau. All integrals at the clipping interface converge because $\sigma < 1$ and the function is locally Lipschitz.

6.4. **Sliding and exhaustion.** For all sufficiently large λ one has $a_\lambda < S$ everywhere. If the value is clipped it equals $-1 < S$, and where the subtraction is positive and the value is unclipped the order follows from $V_\lambda \le S$. If $\zeta = 0$, then $|d_\lambda| \le 2T$. Take λ so large that $2T \le \delta\lambda$. In the compact region $|P| \le 2L_1\lambda$, (6.2) gives the strictly ordered normal profile. Beyond that region the nearest cap point gives the exact exterior tube representation, as in the proof of (6.7). The strict order of the transported saddle then gives $a_\lambda < S$ there as well.

For P in any fixed compact set, P/λ tends uniformly to zero. By the choice of δ, its points satisfy $d_\lambda(P) < -2\delta\lambda$ and $|P| < L_1\lambda/2$ for large λ. Hence $V_\lambda = -1$ there and clipping gives $a_\lambda = -1$.

For a positive witness, fix a sufficiently large λ_0 and take $p_0 = (0, \lambda_0 e_1)$ on the cap axis, with e_1 a unit vector in the y-block. Its inward normal is $(0, e_1)$. Choose a small distance $d_* \in (0, \min(T, \delta\lambda_0))$ along this normal, and denote the resulting point by P_0. It has $|P_0| < L_1\lambda_0/2$ for d_* small, so $V_{\lambda_0}(P_0) = S(r_{\lambda_0}(p_0), d_*) > 0$. The correction vanishes there, hence $a_{\lambda_0}(P_0) > 0$.

Minimize with exterior datum S on B_R^8, and denote the result by v_R. Lemma 2 applies. For the slide we also need continuity up to the boundary. Set $w = v_R - S$, which vanishes almost everywhere outside B_R. Its interior H^s seminorm is finite by the minimizer energy and the smoothness of S. Its cross seminorm is finite by boundedness and the finite cross interaction from Lemma 2. Since w is bounded with bounded support, $w \in H^s(\mathbb{R}^8)$.

By the distributional equations for v_R and S, $Lw = f(v_R) - f(S)$ in B_R, with bounded source. For a compactly supported smooth test, the symmetry argument in Lemma 2 identifies the distributional action with the bilinear kernel pairing in (2.2). Here Cauchy–Schwarz in the H^s seminorm applies to the full pairing because $w \in H^s(\mathbb{R}^8)$. The identity extends to every $H^s(\mathbb{R}^8)$ test vanishing almost everywhere outside B_R. Indeed, for such a test φ, the compression $\varphi(P/(1-\epsilon))$ has support in $\overline{B}_{(1-\epsilon)R}$ and converges to φ in H^s as $\epsilon \downarrow 0$, by smooth density and the uniform norm bounds for dilations near the identity. Mollification with radius smaller than $\epsilon R/2$ then gives smooth interior approximations. The kernel pairing and the source pairing are continuous in H^s, so the weak identity passes to these tests.

The ball satisfies the exterior ball condition, so [16, Proposition 1.1] gives a continuous representative of w on $\mathbb{R}^8$. Since $w = 0$ almost everywhere on B_R^c, continuity gives $v_R = S$ on ∂B_R. An interior contact at P with -1 would give $(v_R + 1)(P) = 0$ and $L(v_R + 1)(P) = f(-1) = 0$. The contact assertion in Lemma 3 would force $v_R = -1$ almost everywhere, contrary to the exterior datum. Hence $v_R > -1$ on $\overline{B}_R$.

For fixed R, define $\mathcal{M}_R(\lambda) = \max_{\overline{B}_R}(a_\lambda - v_R)$. Joint continuity and compactness give continuity of $\mathcal{M}_R$. It is negative for sufficiently large λ, since then $a_\lambda = -1$ on $\overline{B}_R$ and $v_R > -1$ there. If $\mathcal{M}_R(\lambda_0) \geq 0$, there is a scale with $\mathcal{M}_R(\lambda) = 0$. Boundary continuity and $a_\lambda < S$ put the contact point in B_R. The global difference has a maximum of zero there, because $a_\lambda < S = v_R$ on the exterior. Thus $L(a_\lambda - v_R) \geq 0$ at contact. The contact value is $v_R > -1$, so the strict subsolution inequality and equality of the reactions give $L(a_\lambda - v_R) < 0$, a contradiction. This proves $a_{\lambda_0} \leq v_R$. Let $\iota(x, y) = (y, x)$. The transformed function $P \mapsto -v_R(\iota P)$ is a minimizer with exterior datum S, because the energy is invariant under block swap and negation and $-S(\iota P) = S(P)$. Applying the lower bound to it gives

$$a_{\lambda_0}(P) \leq v_R(P) \leq -a_{\lambda_0}(\iota P).$$

Each coordinate reflection sends v_R to another minimizer with the same datum. They are comparable by Lemma 3, and applying the involution reverses the order, giving equality.

The uniform interior estimates of Lemma 2 give a subsequence as $R \to \infty$ converging smoothly on each compact set to v, by diagonal extraction. For any bounded Lipschitz open set A, eventually $A \subset B_R$. Restriction gives minimality of v_R on A, and the closure assertion of the same lemma gives minimality of v on A. Thus v is an entire minimizer. The first variation and interior regularity in Lemma 2, applied on each ball, give $Lv = f(v)$ pointwise. The two bounds give $v(P_0) > 0 > v(\iota P_0)$. If v attained a well, apply the contact assertion of Lemma 3 to $v + 1$ or $1 - v$. The equation makes the Laplacian zero there, which would force v to be constant. The reflection symmetries pass to the limit. This proves Proposition 7.

7. A dimension step by centered slopes

Theorem 8. *Let $m \geq 1$ be an integer. Suppose $v : \mathbb{R}^m \to (-1, 1)$ is an entire minimizer invariant under each coordinate reflection, taken in its smooth representative. Put $c = v(0)$. For every $\gamma \in (0, q'(q^{-1}(c)))$ there is a smooth entire minimizer $U : \mathbb{R}^m \times \mathbb{R} \to (-1, 1)$, invariant under reflections of the first m coordinates, with*

$$\partial_z U > 0, \qquad U(0, 0) = c, \qquad \partial_z U(0, 0) = \gamma.$$

It is nonplanar.

The selection of the centered derivative follows the variational method of Jerison and Monneau [14, Section 4]. In the fractional problem the comparison and the passage to vertical ends both use the finite interaction across a horizontal cut over a bounded base.

7.1. **Cylinder solutions and vertical order.** Write $P = (X, z) \in \mathbb{R}^m \times \mathbb{R}$. Fix $R > 0$, put $B = B_R^m$ and $\mathcal{C}_R = B \times \mathbb{R}$, and for $H > 0$ let $A_H = B \times (-H, H)$. This is a bounded Lipschitz open set. At an edge, in a direction pointing outward through both faces, the cylinder is locally the

subgraph of the minimum of the two smooth graph functions. For $0 \le \tau \le 1$, prescribe on the lateral exterior

$$g_\tau(X,z) = q\big(\tau q^{-1}(v(X)) + (1-\tau)z\big) \qquad (X \in B^c). \tag{7.1}$$

Over B prescribe -1 for $z \le -H$ and 1 for $z \ge H$. The bounded data give a minimizer $u_{\tau,H}$ by Lemma 2. At an interior contact with a well, apply Lemma 3 to $u_{\tau,H}+1$ or $1-u_{\tau,H}$, whose Laplacian vanishes at contact by the equation. The opposite exterior cap has positive measure, so $-1 < u_{\tau,H} < 1$ in A_H.

For $h > 0$ write $a(X,z) = u_{\tau,H}(X,z)$ and $b(X,z) = u_{\tau,H}(X,z+h)$. The shifted function b minimizes in $A_H - he_{m+1}$ by translation. On B^c we have $a \le b$ by monotonicity of g_τ in z. Over B, every point outside the overlap of the two cylinders has either $z \le -H$ or $z + h \ge H$. In the first case $a = -1$, and in the second $b = 1$. Thus the set where $a > b$ is contained in the overlap, up to null sets. Compare a with $\min(a,b)$ on A_H, and b with $\max(a,b)$ on $A_H - he_{m+1}$. The only changed points lie in that overlap, so pairs with a changed endpoint belong to both energy domains. On these pairs the lattice computation (2.3) bounds the summed kinetic integrand of the competitors by that of a, b. This gives finiteness of both competitor energies and a nonpositive summed change. The potential contributions cancel.

If $0 < h < 2H$, then on $X \in B$, $-H - h < z < -H$ we have $a = -1 < b$, since $-H < z + h < -H + h < H$. Pairing this strip with any positive-measure subset of $\{a > b\}$ would give a strictly negative summed change by (2.3) and the positive kernel. This contradicts the sum of the two minimizing inequalities. For $h \ge 2H$ the overlap is empty and the order already holds. Thus for each $h > 0$ the vertical order holds almost everywhere. Continuity gives the order pointwise when both arguments are in the interior.

We call a $[-1,1]$-valued function minimizing in each $A_T = B \times (-T,T)$ a cylinder minimizer. By restriction in Lemma 2, it minimizes on every bounded Lipschitz open set contained in $\mathcal{C}_R$. In particular every vertical translate is a cylinder minimizer. To check minimality of the translate on A_T, compare on the translated slab contained in a taller cylinder.

We will use the following compactness consequence of Lemma 2. Suppose a sequence is $[-1,1]$-valued, minimizes on each fixed A_T eventually, and has lateral values converging almost everywhere on $B^c \times \mathbb{R}$. Uniform interior estimates and diagonal extraction give a locally smooth limit on $\mathcal{C}_R$ along a subsequence. Extend the limit by the limiting lateral values. Convergence then holds almost everywhere on $\mathbb{R}^{m+1}$, and closure on each A_T gives a cylinder minimizer. Any vertical order on the interior passes to the limit by local convergence.

Apply this observation to $u_{\tau,H}$ as $H \to \infty$. Restriction gives the comparisons on A_T for $H > T$, and the lateral datum is fixed. We obtain a nondecreasing cylinder minimizer u_τ with datum g_τ.

7.2. **The end functions.** We first record the kernel bound for pairs across a horizontal cap. For an inside point and a point above A_T, both with base coordinates in B, write the heights as $T-\alpha$ and $T+\beta$, with $0<\alpha<2T$ and $\beta>0$. The lower cap is described by $-T+\alpha, -T-\beta$. For either cap the kernel is $K_{m+1}(X-Y,\alpha+\beta)$. The integral on $B\times B$ and on all $\alpha,\beta>0$ is

$$M_R=\int_0^\infty r\iint_{B\times B}K_{m+1}(X-Y,r)\,dX\,dY\,dr. \tag{7.2}$$

Here the substitution $r=\alpha+\beta$ gives the factor r, since $0<\alpha<r$. For $r\le 1$, enlarging the Y-integral to $\mathbb{R}^m$ and substituting $Y-X=rZ$ bounds the double integral by $C_R r^{-1-\sigma}$. For $r\ge 1$, the bounded base gives $C_R r^{-m-1-\sigma}$. After multiplication by r, both bounds are integrable in their respective ranges because $\sigma<1$ and $m+\sigma>1$. Thus $M_R<\infty$, and the cap interactions in both orientations are bounded by a constant times M_R, uniformly in T.

Now let u be a nondecreasing cylinder minimizer whose lateral data have vertical limits. Translate in z toward either end. These translates are cylinder minimizers, and monotonicity gives a pointwise limit $a(X)$ in the cylinder, independent of z. On the lateral exterior the limit is the corresponding end datum. The compactness observation identifies the extension $(X,z)\mapsto a(X)$ as a cylinder minimizer, with a smooth in B. To show that a minimizes on B in dimension m, we compare energies per unit height.

Write $I_T=(-T,T)$ and set

$$w_T(z,z')=\tfrac12\big(\mathbf{1}_{I_T}(z)+\mathbf{1}_{I_T}(z')\big).$$

For $X,Y\in B$, the ordered pair domain in Q_{A_T} requires $z\in I_T$ or $z'\in I_T$. The weight w_T is bounded by its indicator. For almost every X,Y, the marginal identity (2.4) and Tonelli give

$$\iint_{\mathbb{R}^2} w_T(z,z')K_{m+1}(X-Y,z-z')\,dz\,dz'=2TK_m(X-Y).$$

Indeed, each indicator term integrates to $2TK_m(X-Y)$ before averaging. Thus the weighted $B\times B$ kinetic energy of the lifted a equals $2T$ times the base interior kinetic energy and is bounded by the cylinder energy. For $X\in B$, $Y\in B^c$, the pair domain requires $z\in I_T$ and allows all $z'\in\mathbb{R}$. The marginal identity again gives the factor $2T$, including when the pair is counted in the reverse order. The potential has the same factor. Hence $\mathcal{E}_m(a;B)<\infty$.

Take a finite-energy competitor $b=a$ on B^c. Clipping to $[-1,1]$ fixes the exterior data and decreases energy by Lemma 2. Thus comparison with the clipped competitor suffices. Write b for the clipped function, paste $b(X)$ on A_T, and retain $a(X)$ outside A_T. Call this competitor $\widetilde{b}_T$. The potential difference is $2T$ times its base counterpart. For $X\in B$, $Y\in B^c$, the difference of squared increments is $(b(X)-a(Y))^2-(a(X)-a(Y))^2$ for $z\in I_T$ and all z'. The marginal formula integrates the kernel in z' to $K_m(X-Y)$,

and integrating in z gives the factor $2T$. The reversed pairs give the same factor. For $X, Y \in B$, compare the actual difference of squared increments to

$$w_T(z, z')\big[(b(X) - b(Y))^2 - (a(X) - a(Y))^2\big].$$

When both heights are in I_T, the expressions agree, and when both are outside, they vanish. For pairs straddling a cap, the mismatch is bounded in absolute value by a fixed constant because both profiles take values in $[-1, 1]$. Thus its kernel integral is $O(M_R)$ by (7.2).

The weighted expression, with the $1/4$ kinetic factor, integrates to $2T$ times the base interior kinetic difference. It is absolutely integrable because the base energies are finite. Together with the cap bound, this proves finiteness of the pasted competitor's energy and gives the minimizing inequality

$$0 \le \mathcal{E}_{m+1}(\widetilde{b}_T; A_T) - \mathcal{E}_{m+1}(a; A_T) = 2T\big[\mathcal{E}_m(b; B) - \mathcal{E}_m(a; B)\big] + O(M_R).$$

Divide by $2T$ and send $T \to \infty$ to obtain base minimality.

For $\tau < 1$, the lateral datum g_τ tends to the corresponding well at each end. The constant competitor equal to that well has zero base energy. Nonnegativity of the energy and positivity of the cross kernel then force the end minimizer to equal that constant. Thus the two vertical ends are -1 and 1 for every $X \in B$, using continuity of the interior representatives. For $\tau = 1$, the datum is v at both ends. Lemma 3 identifies each base end minimizer as v, since v is an entire minimizer. Monotonicity between the identical ends gives $u_1(X, z) = v(X)$. This also proves uniqueness among nondecreasing cylinder minimizers at $\tau = 1$.

7.3. **Uniqueness and strict derivative.** Fix $\tau < 1$ and two nondecreasing cylinder minimizers u, w with datum g_τ. For $h > 0$ put $w_h(X, z) = w(X, z + h)$, which is again a cylinder minimizer. The strict lateral order is $u < w_h$, because $1-\tau > 0$ in (7.1). Compare with the minimum and maximum on A_T, retaining the individual exterior values of u, w_h outside A_T. The competitors have finite interior pair energy by the lattice inequality and finite cross energy by boundedness and the cross estimate of Lemma 2. The sum of the potentials is unchanged. Pairs within A_T have nonpositive summed kinetic change. The same holds for pairs from A_T to the lateral exterior, where the exterior values already equal the corresponding minimum and maximum.

If the set $\{u > w_h\}$ had positive measure inside the cylinder, choose a bounded positive-measure part of it. Pair it with a bounded box of positive measure a positive horizontal distance from $\overline{B}$, where $u < w_h$. For every sufficiently large T the first set lies in A_T. The product contributes a finite, strictly positive decrement by (2.3). Both sets and their order differences are fixed, so this gain is independent of T.

The only possible positive error is across the two caps over B. For a cap pair with inside point P, each change at P is at most $|u(P) - w_h(P)|$. The sum of the two kinetic changes is bounded in absolute value before the factor $1/4$ by $8|u(P) - w_h(P)|K_{m+1}(P - Q)$, since the square is 4-Lipschitz

on $[-2,2]$. Use the cap coordinates α,β from (7.2). On the upper cap the inside point has height $T-\alpha$, and for each fixed α the mismatch there tends to zero because u, w_h both have upper end 1. The same holds at $-T+\alpha$ for the common lower end -1. Extend the cap integrands by zero for $\alpha \geq 2T$. Since the mismatch is at most two, a constant times $K_{m+1}(X-Y,\alpha+\beta)$ is an integrable majorant by (7.2). Dominated convergence gives a cap error tending to zero. A fixed strict gain would contradict the two minimizing inequalities, so $u \leq w_h$ almost everywhere, hence pointwise in the cylinder. Letting $h \downarrow 0$ and interchanging u, w gives uniqueness.

The planar minimizer of Section 2 has datum $g_0(X,z) = q(z)$, so uniqueness gives $u_0(X,z) = q(z)$. Each horizontal reflection preserves the cylinder and its energy, preserves g_τ by the symmetry of v, and commutes with vertical translation. Uniqueness therefore gives the horizontal reflection symmetries of u_τ.

If $\tau_j \to \tau$, the lateral values converge pointwise by (7.1). The compactness observation gives a subsequence converging in C^2 on compact subsets to a nondecreasing cylinder minimizer with datum g_τ. Uniqueness identifies it with u_τ, also at $\tau = 1$ by the end argument. Every subsequence admits such a further subsequence, which proves C^2_{loc} continuity in τ.

For $\tau < 1$ and $h > 0$, set $d_h(P) = (u_\tau(P + he_{m+1}) - u_\tau(P))/h$. It is nonnegative in the cylinder and on the lateral exterior by the vertical order and the monotonicity of g_τ. Both arguments of the difference quotient solve the equation in the cylinder, which is invariant under vertical translations. For each h, the quotient is bounded and locally smooth there, so the kernel integral converges absolutely. Subtracting the equations gives $Ld_h = b_h d_h$ there, where $b_h(P)$ is the average of f' along the segment between the two solution values and is bounded uniformly in h.

Suppose $\partial_z u_\tau(P) = 0$. Then $d_h(P) \to 0$. Choose a small closed ball $B_\varepsilon(P)$ compactly contained in the cylinder. For small $h > 0$ the shifted ball is still in a fixed compact subset of the cylinder. On the ball, $d_h(Q) = \int_0^1 \partial_z u_\tau(Q + \theta h e_{m+1})\, d\theta$. The local C^2 bound and the mean value theorem give $|d_h(P) - d_h(Q)| \leq C|P-Q|$ uniformly for small h. The kernel integrand is therefore bounded by $C|P-Q|K_{m+1}(P-Q)$, which is integrable since $\sigma < 1$. Dominated convergence makes the integral on this ball tend to the integral of $-\partial_z u_\tau(Q) K_{m+1}(P-Q)$, which is nonpositive.

Outside the ball, the term $d_h(P) \int_{|P-Q|>\varepsilon} K_{m+1}(P-Q)\, dQ$ tends to zero. The remaining contribution has the sign of $-d_h(Q)$ and is nonpositive. To get a strict bound, choose a compact box G of positive measure in the lateral exterior. On G, the arguments of the smooth profile q for $g_\tau(X,z)$ and for its small vertical translates stay in a fixed compact interval. The continuity of q' there and the mean value theorem give uniform convergence of the difference quotients to

$$\partial_z g_\tau(X,z) = (1-\tau) q'\big(\tau q^{-1}(v(X)) + (1-\tau)z\big) > 0.$$

Write $\mu_G = \min_G \partial_z g_\tau > 0$. For small h we have $d_h(Q) \geq \mu_G/2$ on G, so the contribution from G is bounded above by

$$-\frac{\mu_G}{2}\int_G K_{m+1}(P-Q)\,dQ < 0.$$

It follows that $\limsup_{h\downarrow 0} Ld_h(P) < 0$, contrary to $Ld_h(P) = b_h(P)d_h(P) \to 0$. Hence $\partial_z u_\tau > 0$ throughout the cylinder.

7.4. **Centering and choosing the slope.** For $\tau < 1$, the end values $-1, 1$ and strict vertical increase give a unique h_τ with $u_\tau(0, h_\tau) = c$. At a fixed τ and any $\varepsilon > 0$, the values at $h_\tau - \varepsilon$ and $h_\tau + \varepsilon$ strictly bracket c. Local continuity in the parameter preserves this bracketing for nearby parameters, proving continuity of h_τ on $[0, 1)$. The bracketing also keeps nearby heights in a fixed compact interval. Uniform convergence of the first derivatives there and continuity in height give continuity of

$$D_R(\tau) = \partial_z u_\tau(0, h_\tau).$$

Since $u_0(X, z) = q(z)$, we have $h_0 = q^{-1}(c)$ and $D_R(0) = q'(q^{-1}(c))$.

To determine the limiting centered slope, the heights h_τ themselves may be unbounded. Take any sequence $\tau_j \to 1$ from below and extract a subsequence such that $(1-\tau_j)h_{\tau_j} \to \ell$ in the extended real line. The recentered functions $u_{\tau_j}(X, z + h_{\tau_j})$ are still cylinder minimizers by translation, regardless of the sizes of the heights. Their lateral values are

$$q\big(\tau_j q^{-1}(v(X)) + (1-\tau_j)z + (1-\tau_j)h_{\tau_j}\big).$$

For fixed $X \in B^c$ and z these tend to $q(q^{-1}(v(X)) + \ell)$, where $\ell = \pm\infty$ gives the constant ± 1. The compactness observation gives, along a further subsequence, a nondecreasing cylinder minimizer w with these lateral values and $w(0, 0) = c$. The datum is independent of z, so both vertical ends of w minimize on B with this same datum by the preceding reduction. The comparison function v minimizes on B with exterior datum $v|_{B^c}$ by entire minimality.

If $\ell > 0$, the datum is strictly above v on B^c. Lemma 3 then places the lower end of w strictly above v at $X = 0$, whereas the lower end there is at most $w(0, 0) = c$. If $\ell < 0$, the datum is strictly below v and the upper end is strictly below c at $X = 0$, whereas it is at least $w(0, 0) = c$. Thus $\ell = 0$. Uniqueness with the exterior datum v identifies both end functions with v, and monotonicity gives $w(X, z) = v(X)$. Local C^1 convergence now sends the centered slopes along this subsequence to zero. If $D_R(\tau)$ did not tend to zero, a sequence bounded away from zero would admit such a subsequence, a contradiction.

Since $D_R(0) > \gamma$ and $D_R(\tau) < \gamma$ for τ sufficiently close to 1, continuity gives $0 < \tau_R < 1$ with $D_R(\tau_R) = \gamma$, as in Figure 2. Recenter the corresponding solution by h_{τ_R}. The recentered functions minimize on every bounded Lipschitz open set in their respective cylinders. Each fixed such set in $\mathbb{R}^{m+1}$

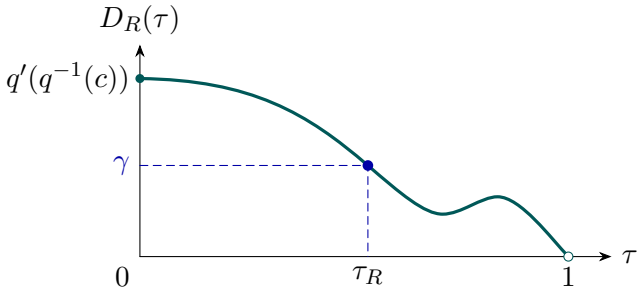


FIGURE 2. At fixed R, the centered slope D_R is continuous on $[0,1)$ and has the planar value $q'(q^{-1}(c))$ at $\tau = 0$. The open circle marks the limit $D_R(\tau) \to 0$ as $\tau \uparrow 1$. The marked crossing of the schematic curve at level γ gives one choice of τ_R by continuity.

lies inside $\mathcal{C}_R$ for large R. The uniform interior estimates and diagonal extraction give a subsequence as $R \to \infty$ converging smoothly on compact subsets to U. Lemma 2 gives entire minimality by closure on each fixed set, hence $LU = f(U)$ on every ball by the first variation and pointwise regularity in that lemma. Local convergence also preserves the horizontal symmetries, $U(0,0) = c$, $\partial_z U \geq 0$, and $\partial_z U(0,0) = \gamma$.

The contact assertion in Lemma 3 gives $-1 < U < 1$, since $U(0,0) = c \in (-1,1)$. The interior estimates of Lemma 2 on translated unit balls bound $\partial_z U$ globally. Lemma 4 gives $L\partial_z U = f'(U)\partial_z U$. If $\partial_z U$ vanished at a point, its Laplacian would vanish there as well. Lemma 3 would then give $\partial_z U = 0$ everywhere, contrary to $\partial_z U(0,0) = \gamma$. Thus $\partial_z U > 0$ everywhere.

If $U(P) = p(e \cdot P)$ for a unit vector e, strict vertical monotonicity on the line $X = 0$ implies $e_{m+1} \neq 0$ and injectivity of p on $\mathbb{R}$. Reflection of a horizontal coordinate X_i and injectivity then give $e \cdot P = e \cdot P - 2e_i X_i$ for all P. Thus $e_i = 0$ for each horizontal index and U is a strictly increasing function of z alone. Its two vertical limits are distinct constants, because its values at any two distinct ordered heights are strictly ordered. Vertical translation preserves entire minimality, and the translates toward each end converge pointwise to the corresponding constant. Closure on each bounded Lipschitz set then makes each constant an entire minimizer.

A constant $a \in (-1,1)$ cannot minimize on every ball. In dimension $n = m + 1$ the competitor equal to 1 on B_R and to a outside has energy difference

$$\frac{(1-a)^2}{2} \int_{B_R} \int_{B_R^c} K_n(P - Q)\, dQ\, dP - |B_R| W(a).$$

The cross integral is finite by Lemma 2 and is $R^{n-\sigma}$ times that for B_1 by scaling. Since $W(a) > 0$, the term $-|B_R|W(a)$ has order R^n and makes the difference negative for large R. Thus the two distinct end values must be

$-1, 1$ in that order. The operator integral for the vertical profile is absolutely convergent because the profile is locally Lipschitz, bounded, and $\sigma < 1$. Fubini and (2.4) now give the one-dimensional equation. Since $\partial_z U > 0$ and the limits are the two wells, one-dimensional uniqueness from Section 2 makes U a translate of q. At the value c its slope must be $q'(q^{-1}(c))$, contrary to the choice of γ. This proves Theorem 8.

Proof of Theorem 1. Use Proposition 7 to obtain v in dimension eight. Since $v(0) \in (-1, 1)$ and the increasing layer q has limits -1 and 1, $q^{-1}(v(0))$ exists. The slope $q'(q^{-1}(v(0)))$ is positive, so half that number is an admissible choice of γ in Theorem 8 with $m = 8$. The resulting smooth entire minimizer U is nonplanar, has $\partial_{X_9} U > 0$, and satisfies (1.1) by the equation for the limit in Theorem 8. □

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